An Economic Constitution of Possibility

Producing Real Freedom, Bounding Contribution Claims, and Revocable Public Power

Jian Gao (高健)

Independent researcher · innersummer@hotmail.com · ORCID 0009-0009-5002-1655

Abstract

Modern economic orders confront three entangled structural dilemmas: nominal income or formal freedom of contract routinely coexists with substantive inability to leave a role; legitimate compensation for exploration, uncertainty-bearing, waiting, and maintenance repeatedly mutates through transfer and rollover into permanent private rule over others' futures; and public coordination oscillates between unaccountable sovereign offices and panoptic surveillance of private life. This paper formulates and proves an economic constitution for an era of rapidly generated possibilities. Its normative core separates **basic participation standing**, **recognized contribution claims**, **scarce-resource requests**, and **revocable governance authority** into four independently justified constitutional entitlements barred from cross-category promotion, while replacing panoptic worthiness testing with **proof-carrying power**: any institution spending common budgets or constraining others' feasible actions must carry verifiable proof of non-overreach, non-displacement of unsettled duties, and zero leakage of private reserves.

The paper gives eighteen formal results with counterexamples in rooted, provenance-preserving ledger models. Theorem 1.2 synthesizes the domain-specific constructions. The Lean proofs of regeneration join six conditions on each actual payment. First, real exit capacity equals maximum matching cardinality across a five-gate bipartite graph; dropping non-capture, payment linkage, or service routing yields three distinct institutional failures with identical scalar value $K = 0$ (the Three Zeros), equal positive capacity $K = 1$ can still conceal missing health provision (the Two Ones), under whole-domain coarse readout $(0, \emptyset)$ the 2×2 diagonal repair matrix forces every post-processing rule to misdiagnose at least one failure branch, original exit demand obeys exact conservation $K + D_{\max}^{\mathrm{orig}} = |C_{\mathrm{act}}|$ (recording 100% of demand as deficit under full capture while keeping capacity and deficit invariant under mere price re-quoting), and the outside-role lifecycle both excludes four retaliation retypings and classifies reappearing former actors as same-debt successors rather than debt-free births (Theorems 2.1–2.2, Proposition 2.3, Theorem 3.1). Second, an open-event innovation law adjudicates two rounds of out-of-menu paid discovery under a mandatory current-living renewal gate (Theorem 4.1); unique issuance provenance bounds cumulative receipts across all slots, transfers, and stages by $0 \le R_o(r) \le Q_o$, ruling out unbounded extraction sequences, while eliminating asymmetric private port exclusion characterizes five-facet absence of capital (Theorem 4.2, Proposition 4.3). Third, single-audit public methods reuse at zero marginal license fee, preserving +1 saved labor unit across every previously lawful continuation and realizing voluntary exit, empty dispatch despite affordable audit, outside production, paid return, and dual `Raw/Ledger` verification along one 25-step trace (Theorems 5.1, 6.1). Fourth, public-function obligations persist across handoffs while expired office jurisdiction vanishes permanently, switching net burden obeys an explicit upper bound, common-cut planning is metered by two invoices across six disjoint ticket phases, and dual-mask dual-replay audit under paid auditor rotation leaks zero information about distinct private reserves across all adaptive strategies,

exogenous noise laws, and timed network channels while `rebuild` reconstructs each participant's exact private root successor (Theorems 7.1–7.4). Fifth, local ledger closure never erases cross-boundary external liabilities, exhausted ledgers obey a two-layer zero-budget squeeze, the paid authority frontier expands strictly at constant instantaneous exit capacity, and one periodic basic payment occurrence simultaneously satisfies all six world-safe regeneration conditions at every depth (Theorems 8.1, 9.1, 9.3, Proposition 9.2).

Keywords: economic constitution; real freedom; real exit capacity; bounded contribution claims; knowledge commons and autonomous time; revocable public power; zero-leakage scoped verification; world-safe regeneration

1 The Constitutional Problem: Ordering Standing, Contribution, and Power When Possibilities Multiply

As automation, reproducible knowledge, and organizational platforms multiply productive possibilities, societies do not automatically become freer. Three sharp institutional paradoxes recur across contemporary economies. First, a person may hold a lawful employment contract or receive a cash transfer on paper, yet lose healthcare, housing access, data portability, or occupational standing the moment an overreaching order is refused—formal consent masks zero substantive exit, and even two regimes with the same positive freedom score may differ in whether basic health security actually exists. Second, exploring unknown methods, bearing uncertainty, waiting across periods, and maintaining shared infrastructure are genuine social contributions without whose compensation open discovery withers; yet once early capital claims are allowed to regenerate principal through secondary transfer, rollover, and legal coding, a finite contribution swells into perpetual private jurisdiction over newcomers [6; 9; 44]. Third, to coordinate large-scale externalities and public services, polities either vest open-ended discretion in permanent offices invoking unfinished emergencies [8; 38] or subject ordinary citizens to panoptic inspection of private life in the name of preventing welfare fraud.

The Freiburg Ordoliberal school recognized early on that an economic constitution is not an external policy add-on to markets, but an integrated legal-economic decision uniting property, debt enforcement, bankruptcy boundaries, and competition order [9; 10]; Buchanan likewise asked why latecomers who never signed an initial compact owe allegiance to inherited rules [21]. Yet constitutional political economy and welfare economics have lacked a **common-ledger, computable formal grammar with paired negative controls** capable of proving when someone can genuinely say no, while accounting under one conservation discipline for risky innovation, commons reuse, expiring public office, privacy-preserving audit, and cross-generational ecological duties. Building on source-native process algebra [1], provenance-preserving conservation calculus [2], canonical society generation [3], and the minimal-permission theory of capital [4], this paper unifies those challenges into one constitutional question: **how can a society simultaneously produce real freedom, settle finite contributions, reuse non-rival knowledge, and bind revocable public power without exporting unaccounted residuals onto vulnerable insiders, external ecosystems, or future generations?**

1.1 Example 1.1: A Three-Minute Pen-and-Paper Audit of the Entire Constitution

Before stating general definitions and theorems, four miniature calculations—verifiable on a single sheet of paper—exhibit the exact mechanics of the paper's four main theorem clusters.

Example 1.1 (Four Pen-and-Paper Constitutional Watersheds). Consider a minimal institutional unit with slots for basic rights, health provision, and learning:

1. **Five-gate real exit, demand conservation, and coarse-metric blindness (Section 2).** Suppose an actor faces candidate continuation actions $C = \{\text{work}, \text{idle}, \text{leave}\}$, where work remains inside the current role, idle stalls without productive continuation, and only leave is a non-idle outside-role target continuation (original exit demand $|C_{\text{act}}| = 1$). Suppose served slots in the current round are $J = \{0\ (\text{rights}), 1\ (\text{health})\}$ (while request 2, learning, waits in the unserved queue), and leave is uncaptured, paid, and connected to slot 0. Maximum bipartite matching size is $K = 1$ with original Hall deficit $D^{\text{orig}}_{\max} = 0$, balancing $K + D^{\text{orig}}_{\max} = |C_{\text{act}}| = 1$. Dropping non-capture, payment linkage, or service routing collapses three distinct institutional pathologies—personal capture, fiscal defunding, and broken infrastructure—into the same scalar $K = 0$ (the Three Zeros), while full capture records 100% of that original exit demand as unmet deficit $D^{\text{orig}}_{\max} = 1$ and mere price re-quoting leaves $(K, D^{\text{orig}}_{\max}) = (0, 1)$ unchanged. Conversely, removing health provision at slot 1 while keeping slot 0 leaves $K = 1$ unchanged (the Two Ones). Even under whole-domain coarse readout $(0, \emptyset)$, the success matrix of `release capture` versus `restore route` across captured and route-broken sources is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, forcing every coarse decision rule to err on at least one branch.
2. **Ex-ante pricing of out-of-menu innovation and origin-capped receipts (Section 4).** Suppose a binary innovation trial has success probability $p = 1/2$, physical failure loss $\ell_{\text{fail}} = 2$ (so probability-weighted uncovered failure loss is $L = (1-p)\ell_{\text{fail}} = 1$), outside option $a = 0$, and agreed risk premium $b = 1/2$. The ex-ante success risk quote is

$$q = \frac{L + a + b}{p} = \frac{1 + 0 + 1/2}{1/2} = 3 \quad (\text{risk cash } +3 \text{ on success, } -2 \text{ on failure, declared expected change } \tfrac{1}{2}(+3) + \tfrac{1}{2}(-2) = +\tfrac{1}{2} = a + b).$$

   Across two fresh rounds $0 \to 1 \to 2$ reaching discovery 2 absent from both quoted menus, each round pays 3 material units and 3 message units (including observation fee 1) while keeping legacy debt 1 unsummed; if current-period living and upkeep go unpaid in either round, the renewal gate blocks the next trial. Meanwhile, if an issuance origin o has cap $Q_o = 10$, unpaid principal plus cumulative receipts satisfy $p_i(B) + R_i(r) = 10$ across arbitrary transfers; whereas permitting free duplicate admission of o in two slots preserves local slot conservation (≤ 10 each) while doubling aggregate extraction to $20 > 10$.
3. **One saved labor unit from commons reuse and non-compulsory dispatch (Sections 5–6).** Suppose auditing a reproducible production method costs 1 labor unit. Once visit 4 pays that audit fee and places the method in the public domain, visit 7 reuses it without re-auditing, leaving +1 extra labor unit across every previously lawful continuation sequence. Along the same 25-step production trace, after the actor exits at step 18 and a peer refuses replacement at step 19, legal audit at step 21 is fully affordable yet actual dispatch remains empty `[]`; at step 22 the exited actor produces +2 net seeds outside the role, and at step 24 voluntarily pays 1 labor unit to return.

4. **Two-invoice planning, zero-leakage scoped audit, and same-occurrence regeneration (Sections 7–9)**. From a common pre-payment cut (with pending prefix length $p = 1$), paying construction Invoice 1 first generates $|\mathcal{L}| = 2$ candidate policies of word length $|w| = 2$, after which selection Invoice 2 bills exact unit count

$$u_{\text{cmp}} = (1 + 1) + 2 \times (1 + 1 + 3 \times 2) = 18,$$

injected disjointly across 6 execution phases so planning can never pretend to be costless omniscience. On the audit side, two branches holding participant 3's private feed reserves of 39 versus 38 (where a 39-unit private transfer succeeds on the first and fails on the second) under a six-public-slot `slot` $\times$ `field` dual mask and dual independent verifiers induce **identical** public observations across every adaptive probing strategy π, exogenous noise law, and timed network channel as audit labor steps down from 20 to 1 across paid handoffs at ordinals 2 and 5, while `rebuild` combines the public post-handoff state with each branch's `ownFrame` to recover its exact private root successor. Finally, at every depth d, one periodic basic payment occurrence p_d simultaneously secures lineage continuity, non-contracting exit, capture-kernel escape, visible residual bearers, open amendment without terminal lock-in, and cross-boundary source accountability.

1.2 Four-Claim Architecture and the Constitutional Possibility Theorem

These four calculations close on a single ledger because the constitution unbundles four legal-economic relations routinely conflated by both absolutist property theory and sovereign state theory, while reversing the direction of surveillance and proof. Drawing on Hohfeld's separation of claim-rights, privileges, powers, and immunities [6], Necula's proof-carrying code paradigm [28], and Nissenbaum's contextual integrity of privacy [32], we state the foundational definition and unifying possibility theorem.

Definition 1.1 (Four-Claim Constitutional Architecture and Proof-Carrying Power). An **accounted economic constitution** over rooted event histories consists of four separated entitlement families, a three-role decoupled payment flow, and a proof-carrying scoped verification operator:

1. **Four independently justified constitutional entitlements (no cross-row automatic promotion)**:
   - *Basic participation standing* $\mathcal{E}_{\text{basic}}$: grounded solely in arrival, living needs, and equal civic standing without prior output or obedience thresholds, certified by five-gate outside-role exit capacity $K > 0$ and structural anti-retaliation;
   - *Recognized contribution claims* $\mathcal{E}_{\text{contrib}}$: grounded in ex-ante or concurrently recognized exploration, uncertainty-bearing, waiting, and maintenance, bound to a unique issuance origin o and cumulative receipt cap Q_o;
   - *Scarce-resource requests* $\mathcal{E}_{\text{scarce}}$: governing rival materials, energy, ecological quotas, and congested time slots under current physical budgets and public allocation rules, never conflated with zero-marginal-copy public methods;

- *Revocable governance authority* $\mathcal{E}_{\text{gov}}$: binding every coordination, licensing, or audit office that alters others' feasible actions to an explicit task, finite term d_{ε}, challenge channel, and post-expiry vanishing $\neg J_{\varepsilon}$. No entitlement implies another: $\mathcal{E}_{\text{contrib}}$ never promotes into $\mathcal{E}_{\text{gov}}$, neither $\mathcal{E}_{\text{contrib}}$ nor debt enforcement may suspend $\mathcal{E}_{\text{basic}}$, and $\mathcal{E}_{\text{basic}}$ never exempts rival consumption from $\mathcal{E}_{\text{scarce}}$.

2. **Beneficiary–provider–budget-bearer decoupling and proof-carrying power (anti-Panopticon)**: Let B_{ben} denote the basic beneficiary, P_{prov} the term-limited service provider, F_{bud} the common budget bearer, and Verify_q the scoped public verifier. Constitutional payment and audit satisfy

$$\begin{aligned} F_{\text{bud}} &\xrightarrow{\text{paid material \& upkeep budget}} P_{\text{prov}} \xrightarrow{\text{delivered service slots}} B_{\text{ben}}, \\ \text{ProofCarrying}(P_{\text{prov}}, q) &\implies \text{ActualSlotPaid}_q \wedge \text{ZeroPrivateLeakage}(B_{\text{ben}}). \end{aligned} \tag{1}$$

   Equation (1) bars both panoptic inversions: neither prior net output from B_{ben} to F_{bud} nor intrusive inspection of B_{ben}'s private reserves may be demanded as a condition of survival; the burden of proof rests on the budget-wielding office P_{prov}.

Theorem 1.2 (The Constitutional Possibility Theorem: Five Constructive Axes and Six Failure Witnesses). In the conservation models of Sections 2–9, with their own physical units and occurrence identities, the constitutional entitlements of Definition 1.1 have the following realizations and failure witnesses.

1. **Five constructive axes**. The respective domain models realize (i) five-gate outside-role exit, complete living portfolios, original exit-demand balance, protection against retaliation, and same-debt reappearance (Theorems 2.1–2.2, Proposition 2.3, Theorem 3.1); (ii) two rounds of paid exploration gated by current living upkeep, the all-slot receipt cap, and the permission characterization of the five forms of capital (Theorems 4.1–4.2, Proposition 4.3); (iii) the commons' $n+1$ net maintenance dividend, the whole-word $+1$ labor dividend, empty dispatch despite affordable auditing, and outside-role production (Theorems 5.1, 6.1); (iv) continuity of public duties, exhaustion of expired jurisdiction, the in-window replacement cost bound, two-invoice billing, and scoped verification and private successor reconstruction in the twin-window model (Theorems 7.1–7.4); and (v) external liability at an empty local ledger, exhausted budgets, strict paid frontiers, and six-factor same-occurrence regeneration (Theorems 8.1, 9.1, 9.3, Proposition 9.2). The periodic regeneration in the fifth axis supplies all six conditions on one payment p_d at each depth.
2. **Six failure witnesses**. Each row gives a tested condition and its observable consequence in the corresponding paired construction.

(a) Three failures and coarse ambiguity	$K(z_c) = K(z_p) = K(z_r) = 0$,
(b) Unpaid current living package	$\text{paidNextTrial?} = \emptyset$,
(c) Two admissions, positive origin cap	$2Q_o > Q_o \quad (Q_o > 0)$,
(d) Closed and open fuel debits	$D_{\text{closed}}(n) - D_{\text{open}}(n) = n + 1$,
(e) Missing payment evidence and extra disclosure	both insufficient, added stock reading $39 \neq 38$,
(f) Empty local ledger and stale payment	$\Lambda_{\text{ext}}(e_{2,k}) \neq \emptyset, \quad p_d \neq p_{d+1}$.

(2)

Proof. The constructions of Sections 2–9 give the domain witnesses in (1). Theorem 9.3 further joins ordinary repair, captured settlement, and actual successor installation on the same periodic payment, which yields its six regeneration conditions. For (2a), Theorem 2.1 gives the Three Zeros. Theorem 2.2 gives two sources with the same $(0, \emptyset)$ readout and different unique effective repairs. For (2b) take the unpaid current package in Theorem 4.1. For (2c) take a positive cap and settle both slots in Theorem 4.2's duplicate-admission example. For (2d) take the two fuel debits in Theorem 5.1. For (2e) use the two insufficient verifications and the distinct twin stocks of Theorem 7.4. For (2f) use the post-settlement family of Theorem 8.1 and distinct successive payments of Theorem 9.3. ■

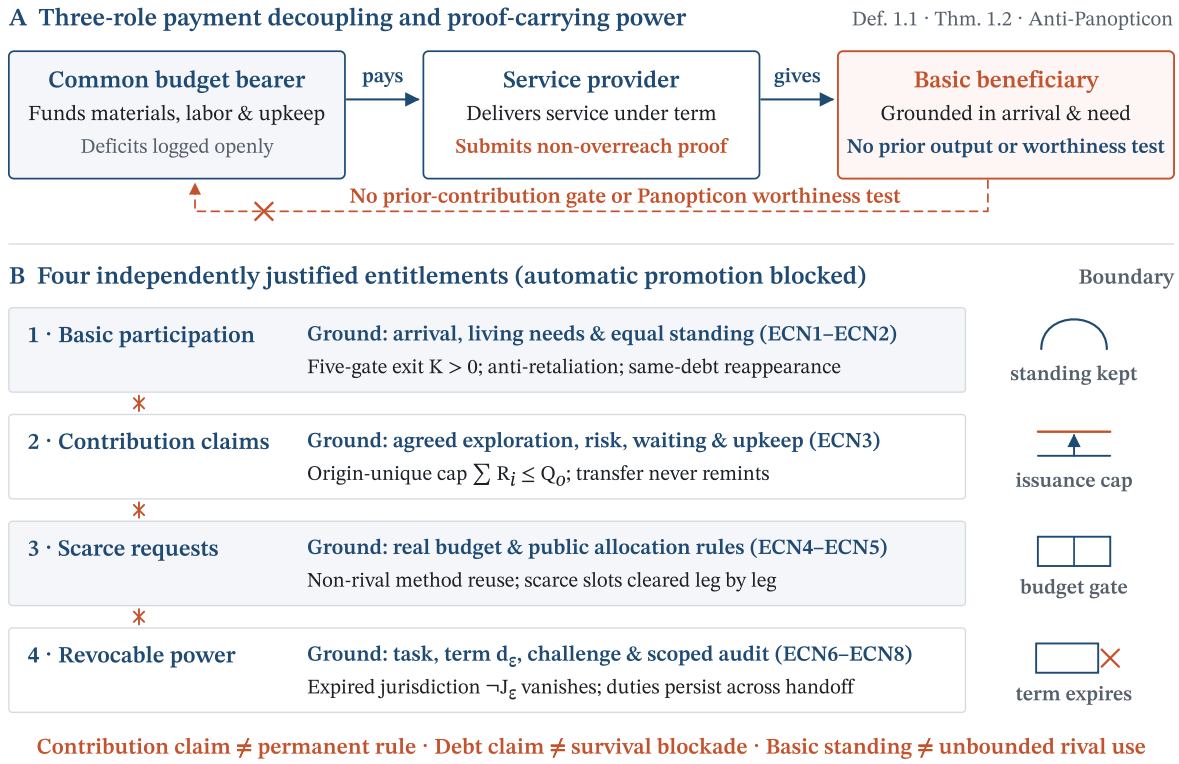


Figure 1 pairs the three-role payment and verification flow of Equation (1) (Panel A) with the firewalls separating the four entitlements (Panel B). Table 1 maps Theorem 1.2 and the eight constitutional principles (`ECN1-ECN8`) to all eighteen formal results and their paired falsifiable negative controls.

Principle & Section	Constitutional Core	Positive Formal Results (Proved in Main Text)	Paired Falsifiable Negative Controls (Drop Condition ⇒ Failure)
Master (Sec. 1) Four claims & proof-carrying power	Separate four entitlements; bind public power to scoped proof	**Thm. 1.2** (Five constructive axes and six failure witnesses)	Equation (2) gives six failures in the corresponding models
ECN1 (Sec. 2) Real freedom & diagnosability	Measure wealth by five-gate exit, full service portfolios, and original demand balance	**Thm. 2.1** (Five-gate matching & Three Zeros); **Thm. 2.2** (Two Ones & coarse obstruction); **Prop. 2.3** (Original demand balance & re-quote invariance)	Dropping any gate yields $K = 0$; full capture records $D_{\max}^{\text{orig}} = \lvert C_{\text{act}}\rvert$ invariant to re-quoting; coarse $(0, \emptyset)$ errs on ≥ 1 branch
ECN2 (Sec. 3) Basic participation & non-employment exit	Survival standing requires no prior output; zero retaliation; shell return keeps debt	**Thm. 3.1** (Four-occurrence three-payment exit equivalence, anti-retaliation, succession & same-debt reappearance)	All four retaliation retypings violate lifecycle invariants; empty-predecessor reappearance never counts as debt-free birth ($\neg$Birth)
ECN3 (Sec. 4) Bounded contribution & open innovation	Admit out-of-menu discovery under current-living gate; cap cumulative receipts per origin at Q_o	**Thm. 4.1** (Open-event law, quote q & two-round trials); **Thm. 4.2** (Origin-unique cap $R_o \leq Q_o$); **Prop. 4.3** (Five-facet no-capital equivalence)	Dropping refunding, observation fee 1, or current living blocks renewal; duplicate admission of o inflates receipts to $2Q_o$; zero law $(0, 0)$ is symmetric yet closed
ECN4 (Sec. 5) Knowledge commons & rival boundaries	Non-rival methods enter public reuse at zero license fee; rival legs and credit settle item by item	**Thm. 5.1** (Zero-fee method reuse & open-replication net fuel dividend $n + 1$)	Closed re-verification spends $n + 4$ fuel leaving baseline reserve 1 (losing $n + 1$ maintenances); credit issuance never counts as completed physical settlement
ECN5 (Sec. 6) Technical progress & autonomous time	Method savings return as autonomous time after net upkeep; capacity to work never implies forced dispatch	**Thm. 6.1** (Whole-word +1 labor dividend, step-21 empty dispatch, outside production & dual audit)	Unnetted gross savings conceal the terminal `CopyReview` supply deficit for the next cycle
ECN6 (Sec. 7) Non-sovereign coordination & zero-leakage audit	Functions persist while offices expire; planning pays two invoices; audit leaks zero private reserve and preserves local `rebuild`	**Thm. 7.1** (Continuous duty & zero sink); **Thm. 7.2** (Switching bound); **Thm. 7.3** (Common-cut two-invoice planning); **Thm. 7.4** (Scoped dual audit, timed zero leakage & `rebuild` root next)	Post-expiry claims never revive J_ε; missing audit payment halts both verifiers; exceeding the dual mask exposes reserve 39 vs. 38
ECN7 (Sec. 8) Cross-boundary duties & bookkeeping	Local balance closure never erases care, ecological, intergenerational, or external duties	**Thm. 8.1** (Non-empty cross-boundary external liability after local open-ledger closure, injective in k)	Treating local zero net balance as global settlement drops live external liabilities $\Lambda_{\text{ext}}(e_{2,k}) \neq \emptyset$
ECN8 (Sec. 9) Open amendment & world-safe regeneration	Amendment stays open without zero-budget refill; each periodic payment secures six regeneration factors	**Thm. 9.1** (Two-layer zero-budget squeeze); **Prop. 9.2** (Strict paid frontier at $K_d = 1$); **Thm. 9.3** (Six-factor same-occurrence regeneration)	Stage relabeling never refills exhausted rows; yesterday's receipt never substitutes for today's payment ($p_d \neq p_{d+1}$)

Table 1: Overview of Theorem 1.2, the eight constitutional principles (`ECN1-ECN8`), eighteen formal results, and paired falsifiable negative controls.

1.3 Plain-Language Glossary and Four Fifteen-Minute Disciplinary Tracks

Six technical terms anchor the formal model in plain institutional concepts: (1) an **occurrence** is an indelible historical event carrying its unique causal prefix (such as one payment, audit, or handoff), rather than a provenance-free closing balance; (2) **five-gate real exit** counts outside-role, non-idle, uncaptured, fiscally paid continuations reaching actually served slots; (3) **issuance origin** o identifies the unique event that first authorized a contribution claim, preserved unchanged across every subsequent transfer; (4) **two-invoice planning** bills policy selection first for enumerating candidate directories and second for evaluating trajectories step by step; (5) **scoped dual-replay audit** restricts public inspection to authorized slot and field masks verified independently along trace and lifecycle routes; and (6) **minimal coface amendment (`U8` / `self-U8`)** resolves rule breakdowns by minimally extending the institutional carrier while preserving live historical obligations.

Readers from distinct disciplines may follow targeted fifteen-minute tracks: **welfare economists and social-policy scholars** can read Sections 1.1–1.2, 2–3, and 6 (Theorems 1.2, 2.1–2.2, Proposition 2.3, Theorems 3.1, 6.1; Figures 2, 4; Table 2); **constitutional lawyers, political philosophers, and corporate-governance scholars** can read Definition 1.1, Theorem 1.2, Section 2.4, Section 3.1, Section 4, and Section 7.1 (Theorems 3.1, 4.1–4.2, Proposition 4.3, Theorems 7.1–7.2; Figures 1, 3); **mechanism designers, monetary theorists, and privacy researchers** can read Section 2.3, Sections 4.1–4.2, 5.1–5.2, and 7.2–7.3 (Proposition 2.3, Example 4.1, Theorems 5.1, 7.3–7.4, Example 7.1; Figure 5; Table 3); and **formal-methods and logic researchers** can pair the main-text proofs with Sections 8–9 (Theorems 8.1–9.3; Figure 6) and Supplement Appendices A–E.

2 Real Freedom and the Grammar of Wealth

An economic constitution must first specify how a society judges whether its members have become wealthier and freer. If the underlying metric counts captured dependence as prosperity or confuses broken service routes with personal idleness, subsequent tax and contract remedies operate on distorted coordinates.

> **Principle 1 (ECN1: Real Freedom and Diagnosable Capability).** A community's primary measure of wealth is neither aggregate monetary output detached from distribution and exit nor a provenance-free scalar freedom score, but the **outside-role real exit routes**, **complete living-service portfolios**, and **actionable failure provenance** actually held by its members without forfeiting basic security or civic standing.

2.1 Five-Gate Real Exit Matching and the Three Zeros

Sen demonstrated that commodity bundles and utility scores cannot stand in for substantive capabilities [22; 40]; Pettit's republican theory established that exposure to another's arbitrary interference constitutes domination even when the master currently stays their hand [29]; Hirschman and Widerquist showed respectively that voice without credible exit degenerates

into supplication and that substantive independence rests on the "power to say no" [18; 35]. Turning those insights into an enforceable constitutional test requires an exact combinatorial characterization of genuine exit.

Let an institutional state s specify a finite set of candidate continuation actions $C(s)$ and a finite set of actually served slots $J(s)$. For $c \in C(s)$ and $j \in J(s)$, let five gate predicates record whether c exits the current dependent role ($R_s(c)$), avoids idle stalling ($\neg I_s(c)$), escapes re-capture by the former controller ($\neg H_s(c)$), carries an actual paid fiscal link ($P_s(c)$), and reaches served slot j along an intact institutional route ($T_s(c,j)$). Define the feasible outside-exit edge set

$$E(s) = \big\{(c,j) \in C(s) \times J(s) \;\big|\; R_s(c) \wedge \neg I_s(c) \wedge \neg H_s(c) \wedge P_s(c) \wedge T_s(c,j)\big\},$$

where each continuation action uses at most one edge and each served slot hosts at most one independent exit edge at a time.

Theorem 2.1 (Five-Gate Real Exit Characterization and Irreversibility of the Three Zeros). For every finite continuation and served-slot state s, outside-role real exit capacity $K(s)$ is uniquely determined by maximum one-to-one matching cardinality on $(C(s), J(s), E(s))$:

$$K(s) = \max\big\{|M| \;\big|\; M \subseteq E(s),\ \forall (c_1,j_1) \neq (c_2,j_2) \in M,\ c_1 \neq c_2 \wedge j_1 \neq j_2\big\}, \qquad K(s) > 0 \iff E(s) \neq \emptyset. \tag{3}$$

Moreover, relative to the canonical positive witness $s_\star$ (with three requests, two served slots, and $K(s_\star) = 1$), constructing the captured state z_c (dropping $\neg H$), the unpaid-link state z_p (dropping P), and the broken-route state z_r (dropping T) yields three pairwise distinct failure origins ($\mathrm{origin}(z_c), \mathrm{origin}(z_p), \mathrm{origin}(z_r)$ distinct) whose scalar exit capacities collapse identically:

$$K(z_c) = K(z_p) = K(z_r) = 0, \qquad \nexists\, \psi : \mathbb{N} \to \{\text{capture}, \text{unpaid}, \text{route}\} \text{ such that } \psi(K(x)) = \mathrm{origin}(x)\ \forall x \in \{z_c, z_p, z_r\}. \tag{4}$$

Proof. Because any matching $M \subseteq E(s)$ consists of edges with pairwise distinct left and right endpoints, $E(s) = \emptyset$ forces $M = \emptyset$ and $K(s) = 0$; conversely, any $(c_\star, j_\star) \in E(s)$ forms a valid singleton matching $\{(c_\star, j_\star)\}$, giving $K(s) \geq 1 > 0$ and proving Equation (3). In $s_\star$, $C(s_\star) = \{\text{work}, \text{idle}, \text{leave}\}$: work fails R, idle fails $\neg I$, and leave alone satisfies $R \wedge \neg I \wedge \neg H \wedge P$ and connects to slot 0, so maximum matching size is $K(s_\star) = 1$. In z_c, z_p, z_r, work and idle still fail $R \wedge \neg I$ while leave violates $\neg H$, P, and $T(\text{leave}, \cdot)$ respectively, leaving $E(z_c) = E(z_p) = E(z_r) = \emptyset$ and hence $K(z_c) = K(z_p) = K(z_r) = 0$. Any putative left inverse ψ can assign only one value to $\psi(0)$, contradicting at least two of the three distinct origins in Equation (4). ■

2.2 Equal-Capacity Blind Spots and Coarse-Readout Repair Impossibility

Theorem 2.1 shows that three opposite institutional pathologies hide behind zero exit. One might hope that ensuring positive capacity $K > 0$, or reporting whole-domain outcome tables across housing, budgets, and service slots, would suffice to guide institutional repair. Arrow's impossibility theorem long ago exposed the informational limits of ordinal preference aggregation [14], while Hayek praised compressed signals such as prices precisely because they discard concrete causal histories [13]. Our second theorem pinpoints the exact constitutional cost of compressed readouts:

positive scalar capacity misses missing essential services, and even whole-domain reachable readout fails completely at selecting institutional repairs.

For each depth $d \in \mathbb{N}$, let $P(x)$ denote the complete living-portfolio predicate (requiring both basic-rights slot 0 and health-service slot 1 to be connected), let $\rho_d(x) = (K(x), \mathrm{Vis}_d(x))$ denote whole-domain coarse readout (exit capacity paired with all visibly served domains), and let $\mathcal{R} = \{\mathsf{rel}\ (\text{release capture}), \mathsf{rst}\ (\text{restore route})\}$ denote candidate institutional repairs with success indicator $\mathrm{Succ}_d(x, r) \in \{0, 1\}$.

Theorem 2.2 (Equal-Capacity Blind Spot and Coarse-Readout Repair Impossibility). For every depth $d \in \mathbb{N}$:

1. **Equal positive capacity ("Two Ones") blind spot**: There exist same-depth institutional sources x_d, y_d sharing identical positive exit capacity while x_d delivers the complete service portfolio and y_d lacks health provision:

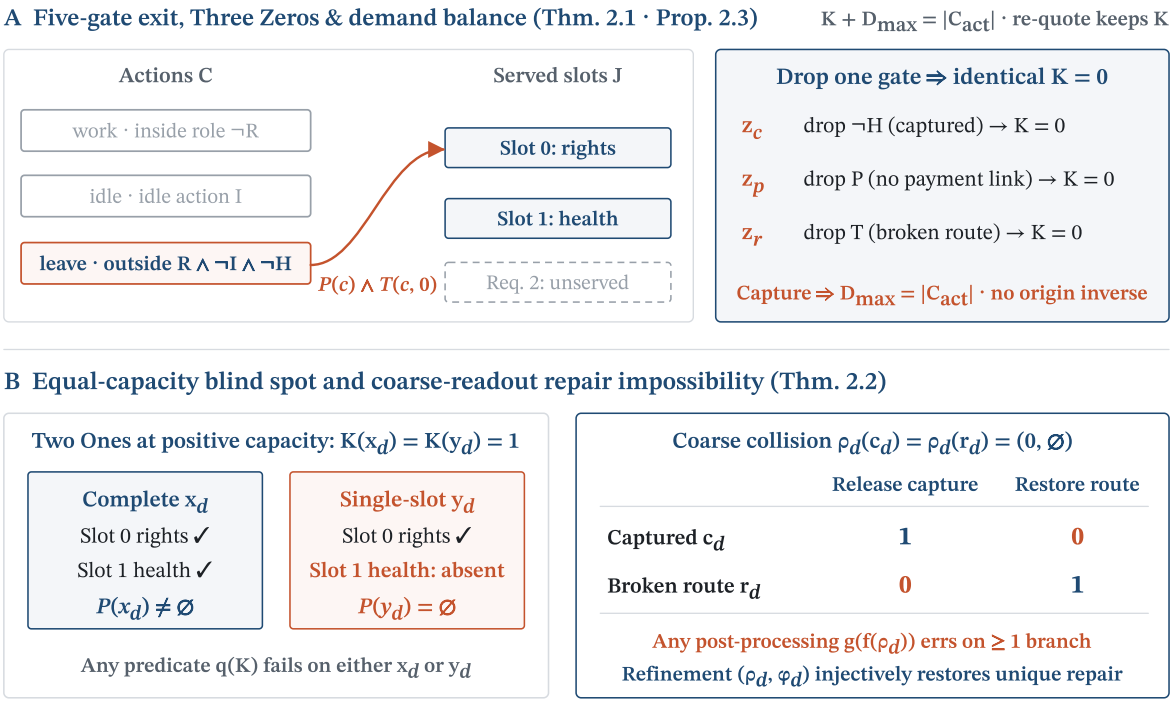
$$K(x_d) = K(y_d) = 1, \qquad P(x_d) \neq \emptyset, \qquad P(y_d) = \emptyset, \tag{5}$$

   so no scalar predicate $q : \mathbb{N} \to \{0, 1\}$ can simultaneously classify $P(x_d)$ and $P(y_d)$ correctly from $K(\cdot)$ alone.
2. **Whole-domain coarse repair obstruction and minimal injective refinement**: There exist a captured source c_d and a broken-route source r_d colliding on whole-domain coarse readout $\rho_d(c_d) = \rho_d(r_d) = (0, \emptyset)$ whose repair effectiveness matrix under $\mathcal{R} = \{\mathsf{rel}, \mathsf{rst}\}$ is strictly diagonal:

$$\begin{pmatrix} \mathrm{Succ}_d(c_d, \mathsf{rel}) & \mathrm{Succ}_d(c_d, \mathsf{rst}) \\ \mathrm{Succ}_d(r_d, \mathsf{rel}) & \mathrm{Succ}_d(r_d, \mathsf{rst}) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{6}$$

   Consequently, for every compression map f and every post-processing rule g, the composite prescription $\hat{r}(x) = g(f(\rho_d(x)))$ fails on at least one branch: $\min_{x \in \{c_d, r_d\}} \mathrm{Succ}_d(x, \hat{r}(x)) = 0$. Restoring failure-kind provenance $\varphi_d(x) \in \{\mathrm{capture}, \mathrm{route}\}$ renders $\tilde{\rho}_d(x) = (\rho_d(x), \varphi_d(x))$ injective across failure branches and uniquely recovers $r^\star$ satisfying $\mathrm{Succ}_d(x, r^\star(\tilde{\rho}_d(x))) = 1$.

Proof. (1) At depth d, let x_d serve both slot 0 (rights) and slot 1 (health) with leave matched to slot 0, and let y_d truncate maintenance to slot 0 alone while preserving the leave-to-0 edge. By Theorem 2.1, both sources have unique maximum matching $\{(\mathrm{leave}, 0)\}$ and thus $K(x_d) = K(y_d) = 1$; yet $P(x_d) \neq \emptyset$ whereas missing health service forces $P(y_d) = \emptyset$, proving Equation (5). Any Boolean map q takes a single value $q(1)$ and therefore misclassifies either x_d or y_d. (2) In c_d, leave fails $\neg H$; in r_d, leave satisfies $\neg H$ but fails T. Neither reaches any outside slot, yielding $\rho_d(c_d) = \rho_d(r_d) = (0, \emptyset)$. Applying rel clears capture on c_d ($\mathrm{Succ}_d(c_d, \mathsf{rel}) = 1$) without fixing the broken route on r_d ($\mathrm{Succ}_d(r_d, \mathsf{rel}) = 0$), whereas rst restores routing on r_d ($\mathrm{Succ}_d(r_d, \mathsf{rst}) = 1$) without lifting capture on c_d ($\mathrm{Succ}_d(c_d, \mathsf{rst}) = 0$), establishing Equation (6). Since $\rho_d(c_d) = \rho_d(r_d)$, any f, g yields $\hat{r}(c_d) = g(f(\rho_d(c_d))) = g(f(\rho_d(r_d))) = \hat{r}(r_d) =: \hat{r}_0$; if $\hat{r}_0 = \mathsf{rel}$ then $\mathrm{Succ}_d(r_d, \hat{r}_0) = 0$, otherwise $\mathrm{Succ}_d(c_d, \hat{r}_0) = 0$. Adjoining $\varphi_d(c_d) = \mathrm{capture} \neq \mathrm{route} = \varphi_d(r_d)$ separates $\tilde{\rho}_d(c_d) \neq \tilde{\rho}_d(r_d)$, and setting $r^\star(\cdot, \mathrm{capture}) = \mathsf{rel}$, $r^\star(\cdot, \mathrm{route}) = \mathsf{rst}$ achieves 1 on both branches. ■



Principle 1: Capability metrics must retain failure provenance and repair authority

Figure 2 unites Theorems 2.1–2.2 and Proposition 2.3 below: compressed indicators may rank outcomes yet fail at institutional medicine; treating capture as a routing shortage pours public subsidies into a locked cage while leaving real exit capacity stuck at zero.

2.3 Original Exit-Demand Conservation and Nominal Re-Pricing Invariance

Two subtler accounting illusions recur in welfare and labor statistics. First, when non-compete clauses or platform exit penalties capture outside continuations, administrative tallies that count only surviving active applicants report "zero applicants, zero unmet exit demand." Second, when real purchasing power and served slots stay unchanged, authorities re-index nominal price observers and claim capability has risen [22; 18; 9].

To foreclose both illusions, let $C_{\mathrm{act}}(s) = \{c \in C(s) \mid R_s(c) \wedge \neg I_s(c)\}$ retain all **original non-idle outside-role continuations** prior to capture filtering, define the maximum Hall shortage on the original-demand bipartite graph $G_{\mathrm{orig}}(s) = (C_{\mathrm{act}}(s), J(s), E(s))$ by $D^{\mathrm{orig}}_{\max}(s) = \max_{A \subseteq C_{\mathrm{act}}(s)}(|A| - |N_{G_{\mathrm{orig}}(s)}(A)|)$, and write $\mathrm{reobs}_v(s)$ for the state obtained by replacing the price observer with any quote $v \in \mathbb{Z}$ while keeping underlying payment receipts, capture rules, and routes unchanged.

Proposition 2.3 (Original Exit-Demand Conservation, Full-Capture Deficit, and Re-Observation Invariance). For every finite exit-capacity geometry state s:

1. **Exact demand conservation and full-capture deficit**: Retaining the pre-capture demand carrier $C_{\mathrm{act}}(s)$ preserves the exact feasible paid edge set, so realized exit capacity plus maximum Hall shortage equals full original exit demand; under full price capture ($\forall c \in C_{\mathrm{act}}(s)$, $H_s(c)$), $K(s) = 0$ and 100% of original exit demand remains recorded as exact deficit $D^{\mathrm{orig}}_{\max}(s) = |C_{\mathrm{act}}(s)|$.

2. **Nominal quote re-observation invariance**: Without new purchasing-power or routing receipts, replacing the price observer by any quote $v \in \mathbb{Z}$ leaves both realized exit capacity and original exit shortage strictly invariant:

$$\begin{gathered} K(s) + D^{\text{orig}}_{\max}(s) = \left|C_{\text{act}}(s)\right|, \\ \forall v \in \mathbb{Z}, \quad \left(K(\text{reobs}_v(s)),\, D^{\text{orig}}_{\max}(\text{reobs}_v(s))\right) = \left(K(s),\, D^{\text{orig}}_{\max}(s)\right). \end{gathered} \tag{7}$$

Proof. (1) Every edge $(c, j) \in E(G_{\text{orig}}(s))$ satisfies $R_s(c) \wedge \neg I_s(c) \wedge \neg H_s(c) \wedge P_s(c) \wedge T_s(c, j)$, so its left endpoint automatically lies in the uncaptured surviving subset; hence $E(G_{\text{orig}}(s)) = E(s)$ and both graphs share maximum matching size $K(s)$. Applying the König–Hall deficiency identity on finite bipartite graph $G_{\text{orig}}(s)$ yields $K(s) + D^{\text{orig}}_{\max}(s) = |C_{\text{act}}(s)|$. Under full capture, $E(s) = \emptyset$ gives $K(s) = 0$ and thus $D^{\text{orig}}_{\max}(s) = |C_{\text{act}}(s)|$ (recording deficit 0 uncaptured versus exact deficit 1 captured on the canonical single-continuation paid root). (2) In $\text{reobs}_v(s)$, each candidate edge retains only payment receipts from the original settlement while the replacement observer contributes zero new payment or routing witnesses, keeping $E(G_{\text{orig}}(\text{reobs}_v(s))) = E(G_{\text{orig}}(s))$ and C_{act} identical and establishing Equation (7). ■

2.4 Dual Constitutional Reach: Binding Sovereign Discretion and Private Government Alike

Theorems 2.1–2.2 and Proposition 2.3 reshape the classical divide between public and private law. Durkheim observed that private contracts presuppose non-contractual social foundations; where bargaining parties stand on grossly unequal survival footing, formal consent merely ratifies forced division of labor [5]. Böhm insisted accordingly that private monopoly and economic coercion enjoy no immunity from public-law scrutiny [9; 10]; Rajan and Zingales showed that regulating access to critical productive resources generates organizational power even without formal title [30]; and Anderson demonstrated that unaccountable workplace hierarchies operate as "private governments" [38]. Recent proposals to replace shareholder value maximization with shareholder welfare maximization [47] or to evaluate rules by ongoing renegotiation [21] leave a decisive constitutional gap exposed by those three results: neither benevolent proxy preferences, nominal price re-indexing, nor renegotiation conducted at $K = 0$ removes domination so long as $\neg H(c)$, $P(c)$, or $T(c, j)$ can be unilaterally severed, leaving full unmet demand $D^{\text{orig}}_{\max} = |C_{\text{act}}|$. An economic constitution therefore applies a **functional criterion of power**: whether dressed as state office, corporate equity, algorithmic platform gatekeeping, or occupational licensing, every power capable of cutting off others' outside-role life paths falls under identical five-gate exit scrutiny, term limits, and contestation rights.

3 Basic Participation and Non-Employment Exit

If real exit is the constitutional yardstick of freedom, on what ground does each member first acquire a viable path outside their current role? That question governs basic participation.

> **Principle 2 (ECN2: Basic Participation Without Prior Contribution Threshold)**. No member forfeits basic subsistence, healthcare, learning, communication, or legal redress for not yet having proved economic output, entered employment, or obeyed an overreaching directive; basic security payments must be decoupled from any single employer's or administrator's retention consent, structurally barring retaliatory retyping of exit into a sanctionable offense.

3.1 Independent Standing, Outside-Role Lifecycle Equivalence, and Structural Anti-Retaliation

Rawls identified the basic structure of society as the primary subject of justice and affirmed equal civic standing prior to market bargaining, yet idealized society as a closed system entered only at birth and exited only at death [19]. Real economic life turns on mid-course exit: quitting an abusive job, declining an unsafe assignment, pausing for caregiving, or switching associations. Whenever basic security flows through current employers or reclassifies job-leavers into punitive "voluntary quit / administrative suspension" categories, nominal rights vanish at the exact moment they are invoked [18; 35; 38].

Fixing historical visit o and source index a, consider each post-settlement lifecycle starting index $i \in \mathbb{N}$ across the four-occurrence, three-payment institutional lifecycle (admission, active operation, outside-role transition, and post-transition review), with complete outside exit-lifecycle type M_i. Let $S_i(c, j)$ denote employer-independent paid service connection of continuation c to served slot j, let $R_i(c)$ denote the full outside-role temporal non-capture contract, let $\mathcal{D} = \{\delta_{\text{pol}}, \delta_{\text{srv}}, \delta_{\text{adm}}, \delta_{\text{emp}}\}$ (policy downgrade, service cutoff, administrative suspension, and employment tie-in) denote the four retaliatory retyping operators, and let $H_{\text{ctrl}} \subset \mathbb{N}$ (finite non-empty) denote the historical roster of non-employment mandate coordinators.

Theorem 3.1 (Basic Participation Lifecycle Equivalence, Anti-Retaliation, Mandate Succession, and Same-Debt Reappearance). Across the four-occurrence, three-payment basic participation and source-owned actor lifecycle:

1. **Outside-role security equivalence**: For every post-settlement starting index $i \in \mathbb{N}$, the outside-role real exit lifecycle M_i is inhabited if and only if an employer-independent paid service edge and an uncaptured outside temporal contract jointly hold:

$$\forall i \in \mathbb{N}, \qquad M_i \neq \emptyset \iff \exists c \in C,\ \exists j \in J,\ (S_i(c, j) \wedge R_i(c)). \tag{8}$$

2. **Exclusion of four retaliation recodings.** In the installed reply to the same refusal, the policy-downgrade, service-cutoff, administrative-suspension, and employment-withholding recodings each fail to carry the original reply authority.
3. **Non-employment term exit and lock-free coordinator succession**: Mandate seats terminate at term without requiring employment status, and successor coordinator identity is

constructively generated by $u_{\text{next}} = \max(H_{\text{ctrl}}) + 1 \notin H_{\text{ctrl}}$, preventing historical incumbents from locking future succession.

4. **Same-debt succession upon reappearance.** Write ExactReappearance($a_\star$) when a target with empty direct predecessors carries the original lineage and an actual reopening edge from the archived debt to a live target slot. The lifecycle law recognizes succession and excludes debt-free birth:

$$\text{ExactReappearance}(a_\star) \implies \text{Succ}(a_\star) \wedge \neg\,\text{Birth}(a_\star) \wedge \text{SameDebt}(\omega_{\text{arch}}, \omega_{\text{target}}). \tag{9}$$

Proof. (1) For every $i \in \mathbb{N}$, M_i packages an active outside continuation c and served slot j witnessing $S_i(c, j)$ together with the c-indexed temporal contract $R_i(c)$: projecting its fields yields the right-hand side of Equation (8), while assembling a complete feasible edge from $S_i(c_0, j_0)$ and pairing it with $R_i(c_0)$ at the same c_0 reconstructs an element of M_i. (2) For the fixed refusal, compare each recoding with the action and service authority of its installed reply. The four original checks rule out that same authority factorization in each recoding. (3) Since $H_{\text{ctrl}} \subset \mathbb{N}$ is finite and non-empty, $m = \max(H_{\text{ctrl}})$ exists and $u_{\text{next}} = m + 1$ satisfies $h \leq m < u_{\text{next}}$ for all $h \in H_{\text{ctrl}}$, hence $u_{\text{next}} \notin H_{\text{ctrl}}$ assumes pending duties without reviving expired authority. (4) When $a_\star$ has empty immediate predecessor together with framework reappearance authority referencing historical identity, left injection yields target succession Succ($a_\star$); any fresh-origin birth witness requires absence of historical identity, contradicting the domain reappearance witness and proving $\neg\,\text{Birth}(a_\star)$; and reopen authority transports the source anchor into a `.live` target slot satisfying SameDebt($\omega_{\text{arch}}, \omega_{\text{target}}$), proving Equation (9). ■

3.2 Same-Ledger Comparison of Basic Income, Universal Basic Services, and Voluntary Job Options

Contemporary social-policy debates pit three flagship proposals against one another: Van Parijs defends unconditional basic income (UBI) as shared employment rent maximizing the minimum of real freedom [27]; Coote champions universal basic services (UBS) in healthcare, education, transport, and care while warning that large cash transfers alone may crowd out public service budgets [45]; and Tcherneva proposes publicly funded, locally administered voluntary job guarantees (JG) to anchor labor standards [41]. Lehner's survey experiment randomized payment amounts for unemployed people in Austria and measured support and stated intentions under income and job guarantees [50].

Theorems 2.1–2.2, Proposition 2.3, and Theorem 3.1 demonstrate that these instruments must be evaluated jointly on one physical and fiscal ledger rather than treated as rival panaceas (Table 2). Cash UBI without real healthcare and care slots J remains vulnerable to monopolistic price extraction or service deserts, producing the Two Ones blind spot of Theorem 2.2 or the route-broken zero z_r of Theorem 2.1 (and Proposition 2.3 confirms that nominal re-pricing never reduces exit shortage); in-kind rationing without portable purchasing power forces individuals to petition administrative windows for every move, inviting capture z_c; and public job options uphold labor dignity **only** as voluntary paid avenues atop unconditional survival security. Turning mandatory work assignment into the gatekeeper of subsistence immediately degenerates into the prohibited

employment retaliation δ_{emp} of Theorem 3.1.

Policy Regime	Portable Cash Link $P(c)$	Real Basic Service Slots $T(c, j)$	Freedom from Capture $\neg H(c)$	Consequence of Refusing Dispatch	Constitutional Diagnosis (Theorems 2.1–3.1)
Cash-only transfer (no service floor)	Present	Fragile (exposed to monopoly pricing or supply deserts)	Depends on debt-enforcement shields	Keeps cash, but affordable health/care slots may be absent	Vulnerable to broken route z_r ($K = 0$) or health-missing Two Ones ($K = 1, P = \emptyset$)
In-kind rationing / discretionary welfare	Weak	Designated items only	Weak (mobility & switching require caseworker approval)	Risks administrative suspension δ_{adm} or downgrade δ_{pol}	Violates proof-carrying Equation (1); prone to capture z_c ($K = 0$)
Conditional workfare	Tied to dispatch	Tied to active compliance	Absent (employer/agency holds retention veto)	Refusal cuts off subsistence ($\delta_{\text{emp}}, \delta_{\text{srv}}$)	Outside-role action set empty; structurally locks $K = 0$
Constitutional participation stack (`ECN1-ECN2`)	Unconditional periodic payment	Co-funded rights, health, learning & care slots	Three-role decoupling + scoped proof	Structural anti-retaliation + voluntary paid work settled separately	Simultaneously secures $K > 0$, full portfolio $P \neq \emptyset$, and zero retaliatory retyping

Table 2: Same-ledger comparison of single-instrument welfare schemes against the constitutional basic participation stack (`ECN1-ECN2`).

4 Settling Contribution, Risk, Waiting, and Open Innovation

Securing unconditional basic participation immediately raises the complementary question: if survival requires no prior output, how does a society reward genuinely risky exploration, intertemporal waiting, and demanding upkeep—and how does it prevent legitimate rewards from hardening once again into perpetual rule over others?

> **Principle 3 (ECN3: Origin-Bounded Contribution Settlement Without Perpetual Rule)**. Genuine exploration, uncertainty-bearing, intertemporal waiting, and maintenance earn explicit compensation under public ex-ante rules, and institutions must remain open to out-of-menu discoveries while paying everyone's current-period living and upkeep first; yet every contribution claim remains bound to its unique issuance origin o and agreed cap Q_o, never regenerating principal across transfers or stage transitions and never converting into perpetual residual control or access veto.

4.1 Open-Event Disposition, Ex-Ante Risk Pricing, and the Current-Living Renewal Gate

Knight separated unmeasurable uncertainty from calculable risk and argued that bearing uncertainty grounds entrepreneurial residual control [7]; Arrow identified the fundamental tension of invention—discovery is risky and costly, yet once produced its optimal utilization price is zero—proposing to decouple research rewards from exclusive usage tolls [15]. Meanwhile, Fraser showed that regimes of capital accumulation routinely subsidize apparent innovation by exhausting uncompensated care and social reproduction [39].

Answering Knight, Arrow, and Fraser on one formal ledger requires three interlocking rules: first, candidate discoveries outside pre-listed menus must be adjudicated by a provenance-based law that never presupposes an enumerated future technology list; second, ex-ante risk quotes must transparently cover principal, waiting allowance, and risk premium without open-ended residual seizure; third, releasing any subsequent trial must pass a strict **current-period living and upkeep payment gate**.

Theorem 4.1 (Universal Open-Event Disposition, Ex-Ante Risk Pricing, and Two-Round Out-of-Menu Exploration). On an open-innovation conservation ledger tracking principal, paid observation messages, and current-period living upkeep:

1. **Universal open-event law and ex-ante risk quote**: For every step t and every candidate method event e (whether inside or outside prior quoted menus), $\mathrm{SourceInnovationLawAt}(t, e)$ decides admissibility without consulting a stored future catalogue. For declared success weight $p \in (0, 1]$, probability-weighted uncovered failure loss $L = \sum_{s \in \mathrm{fail}} w_s \ell_s \geq 0$ (equal to $L = (1-p)\ell_{\mathrm{fail}}$ in a binary trial with failure weight $1-p$ and physical failure loss ℓ_{fail}), outside option $a \geq 0$, and agreed risk premium $b \geq 0$, the unique success-state ex-ante risk quote covering weighted failure loss plus declared expected reward $(a+b)$ is

$$q(L, p, a, b) = \frac{L + a + b}{p}, \qquad \mathbb{E}_{\mathrm{dec}}[\Delta_{\mathrm{risk}}] = p\,q - L = a + b, \tag{10}$$

   where principal repayment, ordinary production income, and rent are excluded from this risk-cash entry; in a binary trial, realized risk-settlement cash equals $+q$ on success and $-\ell_{\mathrm{fail}}$ on failure.
2. **Two-round out-of-menu paid exploration and three negative controls**: There exists a constructive two-round trajectory $0 \to 1 \to 2$ reaching discovery $\mathrm{discovery}(2) \notin \mathcal{M}_0 \cup \mathcal{M}_1$ absent from **both** rounds' quoted menus, in which each round raises fresh financing from empty prior signal, pays 3 material units, 3 message units (including observation fee 1), and every original participant's current living and upkeep bundle, while preserving legacy debt 1 unsummed across distinct units:

$$0 \xrightarrow{\text{material 3, message 3(obs. 1), living paid, old debt 1}} 1 \xrightarrow{\text{fresh funding \& paid living}} \mathrm{discovery}(2) \notin \mathcal{M}_0 \cup \mathcal{M}_1. \tag{11}$$

   Three exact negative controls hold: (i) omitting refunding fails as `unfunded`; (ii) omitting observation fee 1 fails `observation` without `paidSignal`; and (iii) even when future proposals are solvent, leaving current-period living or upkeep unpaid causes `paidNextTrial?` to return `none` and block renewal.

Proof. (1) In risk settlement, success pays $+q$ with weight $p > 0$ while failure branches $s \in \text{fail}$ incur $-\ell_s$ (weighted sum $-L = -\sum_{s\in\text{fail}} w_s\ell_s$). Declared expected net risk change is

$$p\,q - \sum_{s\in\text{fail}} w_s\ell_s = p \cdot \frac{L+a+b}{p} - L = a + b,$$

and since $p > 0$, $pq - L = a + b$ has unique solution $q = (L + a + b)/p$, proving Equation (10). Because SourceInnovationLawAt inspects only submitted resource debits, paid observation receipts, and current-living settlement on the active state, it applies universally to unquoted events e. (2) Executing the finite interpreter across rounds 1 and 2 verifies debits of 3 material units and 3 message units (2 operational plus 1 mandatory observation fee) alongside full living/upkeep settlement and preserved legacy debt 1, landing at technology discovery(2) $\notin \mathcal{M}_0 \cup \mathcal{M}_1$ in Equation (11). Deleting second-round financing violates the material debit guard (`unfunded`); deleting observation fee 1 leaves the observation receipt empty (`paidSignal` absent); and leaving current living unpaid falsifies the first-period living-bundle guard in the renewal check, forcing `paidNextTrial?` to block round 2. ■

Example 4.1 (Risk Quote and Non-Predatory Renewal Entries). In the verified scenario, equal weights $p = 1 - p = 1/2$ and physical failure loss $\ell_{\text{fail}} = 2$ give weighted failure loss $L = \frac{1}{2} \times 2 = 1$; setting outside option $a = 0$ and risk premium $b = 1/2$ in Equation (10) yields $q = (1 + 0 + 1/2)/(1/2) = 3$. Risk settlement selected from the same inventory pays cash $+3$ on success and records -2 on failure, with declared expected change $\frac{1}{2}(+3) + \frac{1}{2}(-2) = +\frac{1}{2} = a + b$. Crucially, however attractive $q = 3$ may look, the current-period renewal gate forbids financing those 3 material units by deferring caregivers', maintainers', or participants' current living bundles, ending innovation subsidized by unpaid social reproduction.

4.2 Unique Issuance Provenance and the All-Slot Cumulative Receipt Cap

Once risk quote q is admitted as a claim, what stops legal coding—priority, durability, universality, and convertibility [44]—from turning it through secondary transfers and rollovers into an inexhaustible rent stream? Ellerman's mathematical treatment of double-entry bookkeeping shows that scalar net balances conceal multi-leg provenance [37]. Our constitutional mechanism stamps every admitted contribution claim with its immutable **issuance origin** o, ensuring that transfers move remaining principal without ever reminting it.

Fix a starting ledger A of issued slots. Let Q_o be the nonnegative rational cap assigned to issuance occurrence o. For an actual finite suffix $r : A \rightsquigarrow B$, let $I_o(A)$ be the slots of origin o in the starting ledger, p_i the balance of slot i, and $R_i(r)$ the cumulative receipt paid on it across holders. Write $R_o(r) = \sum_{i\in I_o(A)} R_i(r)$.

Theorem 4.2 (Unique Issuance Provenance and All-Slot Cumulative Receipt Cap). In the rooted, strictly ordered contribution ledger:

1. **Conservation in the original slot and total receipt cap.** At most one issued slot belongs to any original occurrence o. Payment and holder transfer preserve that slot, and $p_i(B) + R_i(r) = p_i(A) \le Q_o$. Each actual suffix therefore satisfies

$$0 \le R_o(r) = \sum_{i \in I_o(A)} R_i(r) \le Q_o. \tag{12}$$

2. **Unbounded extraction and duplicate-admission control.** No lawful continuation pays an entire rational sequence $d(k)$ unbounded above on one origin. The original free-admission control puts the same origin in two distinct slots, breaking uniqueness. If both slots could pay a positive Q_o in full, local conservation would allow their aggregate to reach $2Q_o > Q_o$.

Proof. (1) Strict issuance order makes $I_o(A)$ empty or a singleton. Paying Δ deducts it from the same slot, and transfer changes its holder. Telescoping the actual suffix gives $p_i(B)+R_i(r) = p_i(A)$. Summing the nonnegative receipts over $I_o(A)$ yields Equation (12). (2) Choose $k_\star$ with $d(k_\star) > Q_o$. Equation (12) precludes that receipt. Under free admission the two distinct slots can share an origin. At full payment of $Q_o > 0$ on each, each local identity is $0 + Q_o = Q_o$ while the sum is $2Q_o$. ■

4.3 Separating Productive Coordination from Perpetual Access Exclusion

Coase explained firms as transaction-cost-saving coordination [11]; Alchian and Demsetz argued that metering team production requires bundling residual claiming, input monitoring, central contracting, membership alteration, and alienation of the firm in one owner [20]; and Grossman, Hart, and Moore defined ownership as residual control over uncontracted contingencies, above all **the power to exclude collaborators from productive assets** [24; 26; 30].

Theorems 4.1 and 4.2 already unbundle paid observation and capped contribution claims ($R_o \le Q_o$). Connecting directly to the companion structural theorem of capitalism [4], we obtain the exact constitutional equivalence characterizing when five-facet capital domination disappears. For law L, state x, port p, and participant u, let $\Phi_L(x, p, u)$ indicate asymmetric private exclusion (controller admitted while u is excluded at the same technical state), let NoCapital(L) denote simultaneous absence of private-control phase, capital-relation kernel, ownership capture, surplus seizure, and newborn historical control, and let NoAsymmetry(L) $\equiv \forall x, p, u,\ \Phi_L(x, p, u) = 0$.

Proposition 4.3 (Five-Facet Absence of Capital via Symmetric Port Access and Trivial Closed Control). For every institutional production law L with decidable participant, good, and instrument equality:

$$\forall L, \qquad \text{NoCapital}(L) \iff \text{NoAsymmetry}(L), \tag{13}$$

The uniformly closed permissions $L_{0,0}$ and the uniformly open permissions $L_{1,1}$ both satisfy the equivalence on the original law's technical, budget, and settlement data.

Proof. The first of the five exclusions rules out every private-control witness and gives $\Phi_L \equiv 0$. A witness for any of the other four facets carries a private-control witness, so $\Phi_L \equiv 0$ excludes

all five and proves Equation (13). Setting `allowed` uniformly false or uniformly true gives the two permissions. With closed permissions the controller cannot enter. With open permissions everyone is admitted, so both satisfy `NoAsymmetry`. Under open permissions, production still proceeds through the original resources, technology, request policy, and paid gates. ■

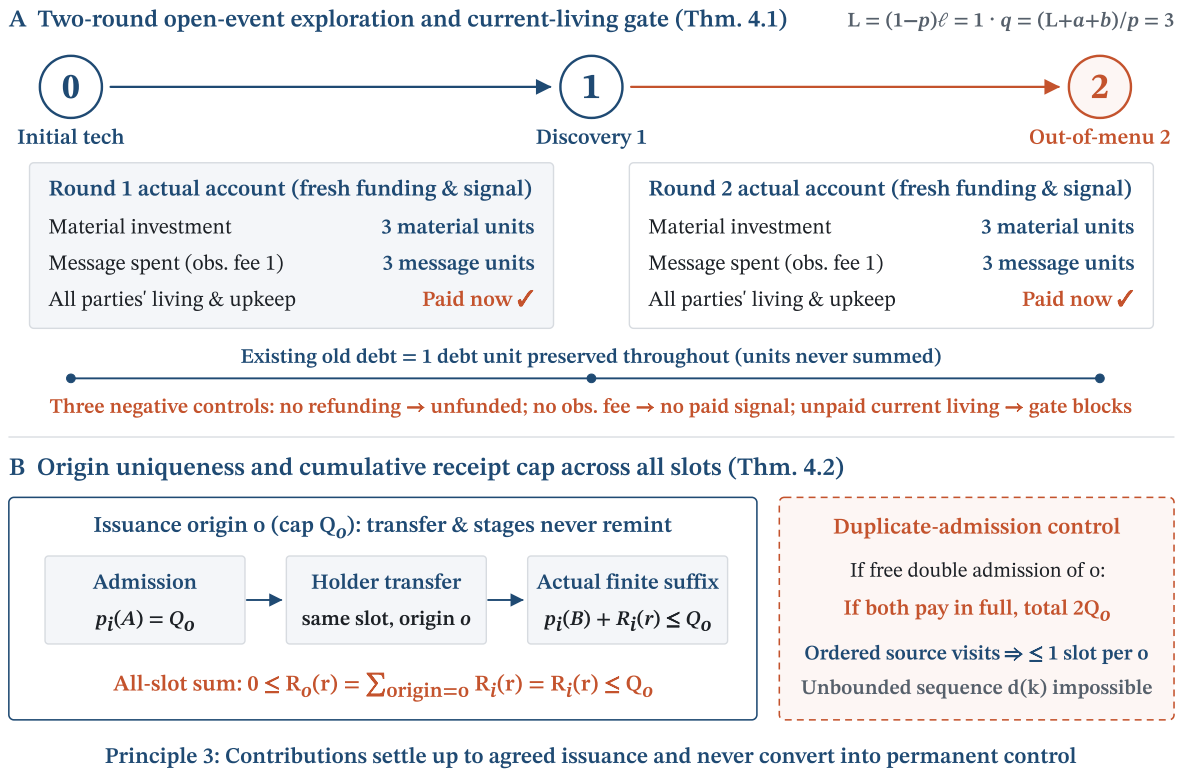


Figure 3 summarizes Section 4: Panel A traces out-of-menu discovery $0 \to 1 \to 2$ gated by current living payments, while Panel B exhibits the telescoping cap $R_o \le Q_o$ alongside the $2Q_o$ duplicate-admission control. Productive coordination thrives without locking collaborators outside the gates.

5 Knowledge Commons and Rival Material Boundaries

Once recognized contributions settle up to Q_o, validated methods, standards, and designs enter the non-rival knowledge commons. Yet egalitarian experiments founder whenever they conflate **zero-marginal-copy information** with **rival materials, energy, maintenance labor, and congested slots**.

> **Principle 4 (ECN4: Separation of Non-Rival Knowledge Commons from Rival Physical Boundaries).** Every non-rival method, standard, or verification procedure whose agreed contribution claim has settled—or that was publicly produced—enters the commons for zero-license-fee reuse; yet every material input, energy draw, ecological quota, congested slot, and maintenance hour consumed in replication must be debited leg by leg in its own physical unit, never disguising material depletion as free knowledge or credit rollover as completed delivery.

5.1 Zero-Marginal-Fee Method Reuse and the Open-Replication Fuel Dividend

Arrow and Benkler showed that non-rival information and modular peer production suffer deadweight loss whenever artificial exclusion tolls block reuse [15; 31], while Ostrom demonstrated that sustainable commons governance separates shared knowledge rules from rival resource-withdrawal limits [33]. Our next theorem locks both sides onto one conservation ledger: **commons reuse yields a strictly growing physical fuel and maintenance dividend while keeping every seed, fuel, and upkeep debit exact**.

Theorem 5.1 (Zero-Marginal-Fee Method Reuse and Open-Replication Net Fuel Dividend). In a conservation-respecting public production system tracking method verification, seed inputs, fuel expenditure, and infrastructure maintenance across replication scale $n \in \mathbb{N}$:

1. **Exact fuel and maintenance dividend over closed re-verification**: From common initial fuel endowment $n+5$ delivering $n+1$ service units plus 1 baseline maintenance in window 1, closed exclusive replication debits $D_{\text{closed}}(n) = n+4$ fuel units (retaining post-window stock $F_{\text{closed}}(n) = 1$), whereas open commons reuse `OpenReplication(n)` debits $D_{\text{open}}(n) = 3$ fuel units (retaining post-window stock $F_{\text{open}}(n) = n+2$); in visit 2, the maintenance budget rule $\lfloor F-1 \rfloor$ (preserving 1 baseline reserve unit) converts that exact fuel saving into $M_{\text{extra}}(n) = n+1$ additional physical maintenances:

$$\forall n \in \mathbb{N}, \qquad \Delta F(n) = D_{\text{closed}}(n) - D_{\text{open}}(n) = F_{\text{open}}(n) - F_{\text{closed}}(n) = n+1 = M_{\text{extra}}(n) \geq 1. \tag{14}$$

2. **Orthogonal physical leg conservation**: Across all n replications, seed stocks, operating fuel, and maintenance labor obey separate unit conservation; zero license fees eliminate redundant verification alone and never conjure free seeds or free fuel.

Proof. (1) Starting from fuel $n+5$, closed execution repeats verification fuel debits across replications for a window-1 debit of $D_{\text{closed}}(n) = n+4$, leaving $F_{\text{closed}}(n) = (n+5) - (n+4) = 1$; `OpenReplication(n)` verifies once and reuses the public method for a constant window-1 debit of $D_{\text{open}}(n) = 3$, leaving $F_{\text{open}}(n) = (n+5) - 3 = n+2$ and saving $\Delta F(n) = n+1$. Visit 2 allocates $\lfloor F-1 \rfloor$ to unit-fuel maintenance tasks, yielding $\lfloor 1-1 \rfloor = 0$ extra maintenances on the closed branch versus $\lfloor (n+2)-1 \rfloor = n+1$ on the open branch ($M_{\text{extra}}(n) = n+1$) with equal cumulative two-window fuel debits, proving Equation (14). (2) Seed, fuel, labor, and maintenance occupy orthogonal coordinates of the inventory vector; the reuse operator bypasses only duplicate audit steps while keeping positive material and operating-fuel debits intact. ■

In concrete terms, when $n = 3$ community clinics or cooperative workshops reuse one verified diagnostic or manufacturing protocol from initial fuel 8, proprietary re-certification burns 7 units in window 1 and funds 0 extra equipment overhauls; open commons reuse spends 3 units and channels all $3+1 = 4$ saved units into round-2 physical maintenance, while still accounting for every reagent dose and kilowatt-hour in its native unit.

5.2 Paired Clearing Legs, Credit Hierarchy, and Monetary Provenance

Honest physical accounting across specialized producers requires clearing and credit institutions. Keynes's International Clearing Union emphasized symmetric adjustment duties on both deficit debtors and surplus creditors [12]; Mehrling highlighted the inherent hierarchy separating lower-tier credit promises from final settlement [34]; Desan showed that money is a constitutional governance architecture allocating public costs and liquidity privileges rather than a neutral barter veil [36]; and Ellerman proved that scalar net zero in double-entry accounts does not imply bilateral leg completion [37].

In our paired-leg clearing model (Supplement Appendix A.5 and Appendix D.1), every bilateral exchange splits into distinct obligation slots $j_i \neq j_{\sigma(i)}$: issuing credit records an open liability whose risk bearer remains tied to the unfulfilled leg until native physical delivery closes that exact slot; unilateral fulfillment never erases the counterparty's open obligation. Table 3 summarizes the four-way accounting boundary separating non-rival commons, rival physical budgets, and credit clearing.

Accounting Dimension	Non-Rival Knowledge & Method Commons	Rival Materials, Ecology & Congested Slots	Credit & Monetary Clearing Instruments
Marginal reuse cost	0 license fee after single paid verification	Positive material, energy, or slot consumption per use	Issuing a promise creates zero matter; redemption requires real goods/services
Access & pricing rule	Symmetric open reuse; permanent patent blockade barred	Bound by current physical budgets and public allocation rules	Strictly separates basic participation, capped contribution ($\leq Q_o$), and rival quotas
Transfer & scale effect	Wider reuse expands net fuel/maintenance dividend $n+1$ (Thm. 5.1)	Over-extraction depletes stock or leaves next-round supply gaps	Transfer moves residual principal only; circulation never remints Q_o (Thm. 4.2)
Settlement criterion	Publicly replayable verification receipt	Per-good conservation without cross-unit summation	Each paired leg closes only upon native fulfillment; rollover $\neq$ settlement

Table 3: Four-way constitutional accounting separation across non-rival knowledge commons, rival material boundaries, and credit clearing.

6 Technical Progress and the Autonomous Time Dividend

Even when methods enter the commons, industrial societies face a two-century paradox: why have repeated leaps in productivity left ordinary people trapped between overwork and fear of displacement?

> **Principle 5 (ECN5: Priority of Autonomous Time Dividends and Prohibition of Compulsory Dispatch)**. Socially saved labor from automation and public method reuse—after deducting real material replenishment, infrastructure maintenance, care support, and ecological restoration—must return on a transparent net ledger primarily as **autonomous disposable time** for all members or jointly approved public-service improvements; possessing work capacity or audit affordability never authorizes compulsory labor dispatch.

6.1 Whole-Word Commutation of One Saved Labor Unit and the 25-Step Non-Compulsory Production Trace

Acemoglu and Restrepo's task framework shows that automation generates both labor-displacing substitution effects and output-expanding productivity effects—aggregate output growth never automatically translates into worker welfare or leisure without institutional rules [42]. Empirically, Fan, Schor, and colleagues' study across 141 voluntarily participating organizations (2,896 employees) adopting pay-maintained four-day workweeks, with 12 comparison organizations found reduced burnout and improved health alongside workflow reorganization, while underscoring non-random organizational self-selection and the need to verify that care burdens are not shifted off-book [49]. Making time liberation constitutionally enforceable requires proving two properties on the same formal production carrier: first, saved labor from public method reuse commutes across **every** previously lawful continuation word; second, even when legal audit is fully affordable, an exited member retains **empty compulsory dispatch** alongside **independent outside-role production** [18; 35; 11; 20].

Theorem 6.1 (Single-Audit Method Reuse Time Dividend and Non-Compulsory Production Trajectory). In the public production process with slot births, paid method auditing, role withdrawal, and dual-route verification:

1. **Three equivalent readings and whole-word commutation of one saved labor unit**: Suppose visit 4 pays 1 labor unit to audit and register a reproducible method and visit 7 reuses it without re-auditing. Comparing shared-method terminal state x_{shared} with repeated-audit baseline x_{base}: (i) current-run labor debit on the parent cell decreases by exactly 1 while all credits across all participant addresses and all seven goods match; (ii) full terminal carriers satisfy $x_{\text{shared}} = x_{\text{base}} \oplus 1_{\text{labor}}$; and (iii) every finite action word $w \in \mathcal{W}_{\text{legal}}(x_{\text{base}})$ lawful from x_{base} remains lawful from x_{shared} and preserves that exact unit labor surplus:

$$\forall w \in \mathcal{W}_{\text{legal}}(x_{\text{base}}), \qquad \text{execute}(x_{\text{shared}}, w) = \text{execute}(x_{\text{base}}, w) \oplus 1_{\text{labor}}. \tag{15}$$

2. **Voluntary exit, empty dispatch despite affordable audit, and outside production along one 25-step trace**: Along the constructive 25-step trajectory (boundary states $0 \to 25$, execution steps $0, \dots, 24$), the actor exits at step 18 (reaching state 19) and a peer refuses office at step 19 (reaching state 20), after which:

$$\text{AffordableAudit}(21) = \text{true} \ \wedge \ \text{Dispatch}(21) = [\,], \qquad \Delta\,\text{Seed}_{\text{outside}}(22) = +2, \qquad \text{PaidReturn}(24) = 1_{\text{labor}}. \tag{16}$$

   Thus at step 21 legal audit is affordable yet actual dispatch is empty `[]`; at step 22 the exited actor independently produces outside the role (net +2 seeds while paying living and upkeep); and at step 24 the actor voluntarily pays 1 labor unit to re-enter.
3. **Independent `Raw` and `Ledger` acceptance with honest supply-gap disclosure**: The same 25-step trace independently passes `Raw` occurrence scanning and `Ledger` slot replay across 6 original slots plus 8 newborn slots from 4 births (at visits $4, 7, 9, 10$), while terminal `CopyReview` explicitly preserves the next cycle's material supply deficit rather than masking net shortages behind gross labor savings.

Proof. (1) At visit 7, baseline execution debits 1 audit labor unit whereas shared execution reads visit 4's verified method receipt and skips that debit with all other transitions identical, yielding $x_{\text{shared}} = x_{\text{base}} \oplus 1_{\text{labor}}$. By induction on the length of $w \in \mathcal{W}_{\text{legal}}(x_{\text{base}})$, request admissibility is monotone in non-negative extra labor and ledger updates commute with $+1_{\text{labor}}$, establishing Equation (15) across all natural-number addresses. (2) Direct execution of steps 18–24 sets withdrawal and peer-refusal standing at states 19–20; at step 21, budget guards confirm audit affordability while the non-compulsion guard (requiring active acceptance signature) outputs `Dispatch(21) = []`; step 22 invokes the public method outside the role to net +2 seeds; and step 24 executes signed paid return with labor debit 1, proving Equation (16). (3) `Raw` scans the 25 occurrences directly whereas `Ledger` replays the $6 + 8 = 14$ service/birth slots without consulting `Raw`'s certificate, both returning valid acceptance while terminal `CopyReview` records the exact residual material gap. ■

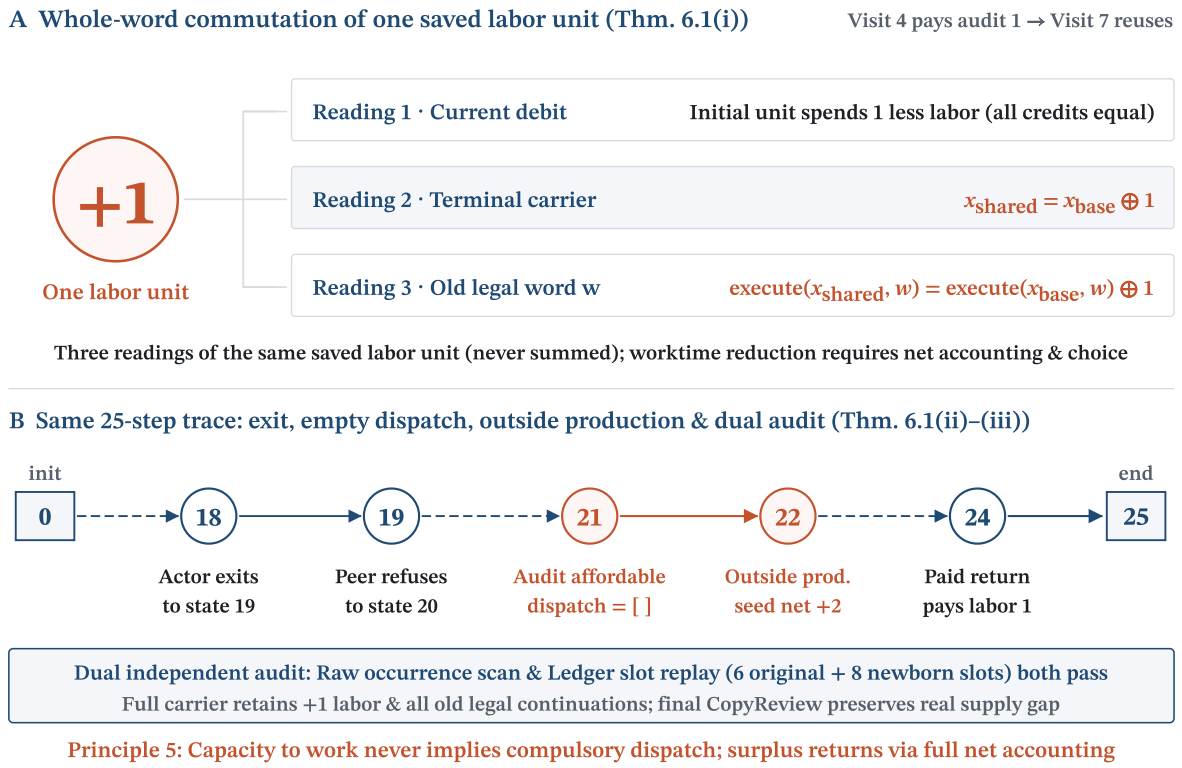


Figure 4 highlights two guardrails: Panel A's three lines are three equivalent views of **one** saved labor unit (never summed to three), and Panel B shows both that fiscal capacity to pay or audit never licenses forced dispatch and that gross time savings must clear net upkeep, care, and replenishment first [39].

7 Non-Sovereign Public Coordination and Anti-Panopticon Scoped Verification

Even with exit, basic security, capped claims, commons reuse, and time dividends in place, large societies still coordinate shared infrastructure, public services, and rule enforcement. Rejecting both Schmittian sovereign exception [8] and Benthamite panopticism, how can public coordination avoid both institutional paralysis and Leviathan capture?

> **Principle 6 (ECN6: Continuous Public Functions Under Expiring Offices, Metered Planning, and Zero-Leakage Scoped Audit)**. Every public coordination, licensing, or audit seat holds task-bound authority limited to an explicit term d_ε: upon expiry its jurisdiction vanishes permanently while unsettled public obligations persist intact across handoffs; public planning must pay two explicit resource invoices for candidate enumeration and evaluation; and public verification uses dual-mask dual-replay audit under paid auditor rotation to certify public slot delivery with zero leakage of private reserves.

7.1 Continuous Public-Function Responsibility, Zero Post-Expiry Jurisdiction, and Switching-Cost Bounds

Sovereignist legal theory routinely invokes unfinished public emergencies to justify indefinite incumbency [8], while Kelsen's pure theory of law stops authorization regress via a presupposed static basic norm [16]. By contrast, Ostrom distinguished collective service provision from replaceable production [33]. Theorems 7.1 and 7.2 formalize this separation: **public obligations persist continuously along the causal track while officeholders' jurisdiction vanishes irrevocably after expiry, and replacing a captured provider incurs at most an explicitly bounded net fiscal cost**.

Theorem 7.1 (Continuous Public-Function Responsibility and Zero Sovereign Absorbing Sink). Along every on-path evolution segment $C_s \to C_{s+k} = \mathrm{causalAt}(C_s, k)$:

1. **Cross-handoff preservation of directory, provenance, and unsettled obligations**: Across arbitrary provider replacements, the request-directory prefix, received opportunity bases, original provenance $\mathrm{origin}(r') = \mathrm{origin}(r)$ of every unsettled obligation (settled = false), and prior financial receipts retain their provenance. Review can continue on the same responsibility path.
2. **Permanent post-expiry vanishing of office jurisdiction (no sovereign sink)**: For every mandate ε with statutory expiry d_ε, once any reachable suffix state x_k passes d_ε, the incumbent's effective jurisdiction remains false across **all** subsequent accepted suffixes:

$$\forall k \geq 0, \qquad \mathrm{clock}(x_k) > d_\varepsilon \implies \neg J_\varepsilon(i, x_k), \tag{17}$$

ruling out any absorbing domain in which unfinished obligations keep expired office authority alive.

Proof. (1) Directory prefixes, financial histories, and unsettled obligation slots attach to the public-function state rather than the incumbent's seat; each step transition updates incumbency pointers while preserving $\text{origin}(r') = \text{origin}(r)$ and prior receipt provenance, which inducts across k to $C_{s+k} = \text{causalAt}(C_s, k)$. (2) Effective jurisdiction is defined by $J_\varepsilon(i, x) \equiv (\text{holder}(x) = i) \wedge (\text{clock}(x) \leq d_\varepsilon) \wedge \text{Active}(\varepsilon, x)$. Because lawful transitions advance clock monotonically (and legal expiry witnesses an all-accepted path crossing d_ε while preserving raw incumbency and responsibility transport), $\text{clock}(x_{k_0}) > d_\varepsilon$ implies $\text{clock}(x_k) > d_\varepsilon$ for every suffix $k \geq k_0$, falsifying $J_\varepsilon(i, x_k)$ permanently in Equation (17). ■

Formal term limits still fail if incumbents engineer prohibitive switching costs [38; 47]. Comparing candidate a against baseline b within the same checked window under price vector p_g, write F_s, F_r for selected and complementary net financial burdens, $Q_g^{\text{before}}, Q_g^{\text{after}}$ for physical inventories, and $B(a; b), G(a; b)$ for gross switching burdens and efficiency gains.

Theorem 7.2 (In-Window Provider Replaceability and Net Capture-Burden Bound). Across every lawful same-window provider comparison:

1. **Dynamic deficit-cell identity and whole-window financial balance**: Every dynamically born obligation cell satisfies remaining + paid = quantity, and selected plus complementary financial outlays equal valued physical inventory change:

$$F_s + F_r = \sum_g p_g (Q_g^{\text{before}} - Q_g^{\text{after}}).$$

2. **Three-way switching verdict and explicit net capture bound**: Evaluation exhaustively returns (i) unpaid baseline service, (ii) a strictly improving executable replacement, or (iii) under computed stability ($\{a \in M_i(b) : U(a) > U(b)\} = \emptyset$), the explicit upper bound for every checked candidate a:

$$F_s(b) - F_s(a) \leq B(a; b) - G(a; b). \tag{18}$$

Proof. (1) Each cell initializes remaining = quantity, paid = 0 and transfers Δ from remaining to paid upon settlement; summing paired transfers across the window cancels internal flows against $\sum_g p_g (Q_g^{\text{before}} - Q_g^{\text{after}})$. (2) Pointwise decomposition $\max(0, c_j) - \max(0, -c_j) = c_j$ yields $B(a; b) - G(a; b) = \sum_j c_j(a; b) = F_r(a) - F_r(b) + V(b) - V(a)$. Under computed stability $U(a) \leq U(b)$ where $U(w) = V(w) - F_s(w) - F_r(w)$, rearranging gives Equation (18) with full gains $G(a; b)$ deducted. ■

7.2 Common-Cut Metered Planning and Six-Phase Disjoint Ticket Slots

Hayek asks how dispersed knowledge enters economic coordination [13]. Dütting and colleagues use constrained learning to find auction allocation and payment rules [43]. Here comparing rules is a paid activity from a common pre-payment cut, with separate invoices for making the catalogue and comparing its entries.

Theorem 7.3 (Common-Cut Adaptive Planning Equivalence, Two-Invoice Billing, and Six-Phase Slot Conservation). From a common pre-payment cut c_0 with pending word length $p \in \mathbb{N}$ and ordered nonempty candidate catalogue $\mathcal{L}$:

1. **Equivalence between finite joint tables and all admissible adaptive opponents**: For every relation Ψ comparing full present-window, future-recovery, and financing tensors, satisfaction across the finite joint table set $\mathcal{T}$ is equivalent to satisfaction against **every** admissible adaptive opponent $\omega \in \Omega_{\mathrm{adm}}$:

$$(\forall t \in \mathcal{T},\ \Psi(t)) \iff (\forall \omega \in \Omega_{\mathrm{adm}},\ \Psi(\omega)), \tag{19}$$

   producing either a non-dominated deployable policy whose installed window equals its compared full initial-cut window (including its own payment prefix) or a per-intent failure row.
2. **Two-invoice catalogue billing and six-phase disjoint slot injection**: Policy comparison requires first paying construction Invoice 1 to enumerate the ordered candidate directory $\mathcal{L}$, followed by paying catalogue-selection Invoice 2 whose exact unit count is

$$u_{\mathrm{cmp}}(p, \mathcal{L}) = (1 + p) + \sum_{w \in \mathcal{L}} (1 + p + 3|w|), \tag{20}$$

   and injects paid work events across six pairwise disjoint execution phases $(e_0, e_1, e_2, e_3, e_4, e_5)$ plus unused balance unused satisfying $\sum_{j=0}^{5} e_j + \mathrm{unused} = u$.

Proof. (1) Fix a comparison horizon within the remaining window. A common adaptive opponent induces a finite joint table at the histories queried by both intents and preserves their full execution values. Conversely, reading each table entry at its query key and using the admitted default elsewhere realizes that table by an adaptive opponent. The two directions give Equation (19). Screening returns the deployable policy with its checked payment prefix, or per-intent failure rows. (2) Once construction Invoice 1 is paid, the catalogue session generates the ordered candidate list $\mathcal{L}$; selection Invoice 2 bills $(1 + p)$ for the unpaid baseline reference at pending prefix length p plus $(1 + p + 3|w|)$ for each candidate word $w \in \mathcal{L}$, summing to Equation (20). The ticket-allocation map injects each paid work event into mutually disjoint phase slots $j \in \{0, \ldots, 5\}$, conserving $\sum_{j=0}^{5} e_j + \mathrm{unused} = u$. ■

Example 7.1 (Two-Invoice Bill for Two Candidate Policies). With pending prefix length $p = 1$, after paying construction Invoice 1 to generate $\mathcal{L} = [w_1, w_2]$ of lengths $|w_1| = |w_2| = 2$, Equation (20) bills selection Invoice 2 for $1 + 1 = 2$ on the unpaid baseline reference plus $2 \times (1 + 1 + 3 \times 2) = 16$ across the two candidates, totaling $u_{\mathrm{cmp}} = 18$. Out of an approved work quota $u = 20$, spending $\sum_{j=0}^{5} e_j = 18$ across the six disjoint phases leaves exact balance unused $= 2$, preventing unbilled search or double-counted tickets.

7.3 Anti-Panopticon Scoped Dual-Replay Verification, Timed Zero Leakage, and Private Successor Reconstruction

How can citizens be certain public funds actually reached intended service slots without building a surveillance state, while still computing their own exact private continuations locally? Proof-carrying verification shifts the burden onto power [28], while Nissenbaum's contextual integrity defines privacy as enforcing boundary-appropriate information flows [32].

On our six-slot twin-window (`WindowTwin`) public wire, all six slots are public service slots, and every public audit query specifies a public `slot` $\times$ `field` mask pair $q \in \mathrm{Fin}\,64 \times \mathrm{Fin}\,4$. Admitted reads supply masked evidence to the trace and lifecycle verifiers while debiting 2 audit-labor units per read, consuming 19 labor units together with mandatory paid auditor handoffs at ordinals 2 and 5 as the audit budget steps down from 20 to 1. Meanwhile, two twin branches $b \in \{\mathrm{false}, \mathrm{true}\}$ hold genuinely distinct private reserves under participant 3 (outside the six public service slots): participant 3's private feed stock is 39 on false (where a 39-unit private internal transfer succeeds) versus 38 on true (where that same 39-unit transfer is rejected). Write $\mathrm{rebuild}(x_{\mathrm{vis}}, f)$ for the operator replacing the private feed coordinate of public state x_{vis} with local frame f while keeping all actors, non-private goods, claims, technology, clock, and round intact.

Theorem 7.4 (Scoped Dual Verification, Timed Zero Leakage, and Exact Private Successor Reconstruction). On the six-slot dual-mask twin-window audit model:

1. **Proof-carrying audit and paid rotation.** For any scoped query $q \in \mathrm{Fin}\,64 \times \mathrm{Fin}\,4$, either verifier's verified verdict entails a paid occurrence at the original slot. When the evidence carries no payment, both return insufficient scoped evidence. The actual reads and paid handoffs at ordinals 2 and 5 consume 19 labor units and revoke the outgoing auditors' further reads.
2. **Adaptive observations and finite independent draws.** Fix the public channel. Each experiment specifies its public initial state, adaptive strategy, horizon, and a finite rational-weight law ν_j on complete exogenous tapes. For the same sequence of public experiments run independently, the complete ordered observations and their joint rational weights agree under any two sequences of private selectors b, b':

$$\begin{aligned} &\mathrm{Verified}_q(x) \Rightarrow \mathrm{ActualSlotPaid}_q(x), \\ \forall \pi, k, \quad &\mathrm{Obs}_k(\pi, \mathrm{false}) = \mathrm{Obs}_k(\pi, \mathrm{true}), \\ &\forall \nu, b, b', \quad \mathbb{P}^b_\nu = \mathbb{P}^{b'}_\nu. \end{aligned} \tag{21}$$

3. **Public timing and private reconstruction at handoff.** In the proved framed channel, any two frames f, f' have the same complete timed observations at a common public initial state, strategy, tape, and horizon. With unit-energy revision cost and a word of energy actions, `rebuild` commutes with actual next. At the original audit handoff, each branch's public post-state and its own source-generated private frame recover its exact root successor:

$$\mathrm{TimedRun}(f) = \mathrm{TimedRun}(f'), \qquad \forall b \in \{\mathrm{false}, \mathrm{true}\}, \quad \mathrm{rebuild}(x^+_{\mathrm{pub}}, \mathrm{ownFrame}(b)) = x^+_{\mathrm{root}}(b). \tag{22}$$

Proof. (1) Scoped evidence is bound to the original slot. Trace acceptance and lifecycle

acceptance each recover its actual payment. With the payment field absent, both return insufficient evidence. The original execution accounts for each paid read, both handoffs, and both revocations. (2) The twin branches hold 39 and 38 units of private reserve. A private 39-unit transfer distinguishes them locally, while permitted projections agree. By induction on the horizon, equal histories yield equal queries and then equal public observations. For each common exogenous tape, channel coupling preserves the complete state and observation sequence. Rational weighting gives equality for a single run. The finite product starts at the unit law on the empty list, then prepends one complete observation and multiplies its weight by that of the tail. The single-run equality and the induction hypothesis preserve every outcome and weight. (3) The framed channel follows one public execution path. Energy actions preserve the private coordinates used by reconstruction. The original handoff seeds satisfy `SeedTwin`, so reconstruction gives each branch its own actual root successor. ■

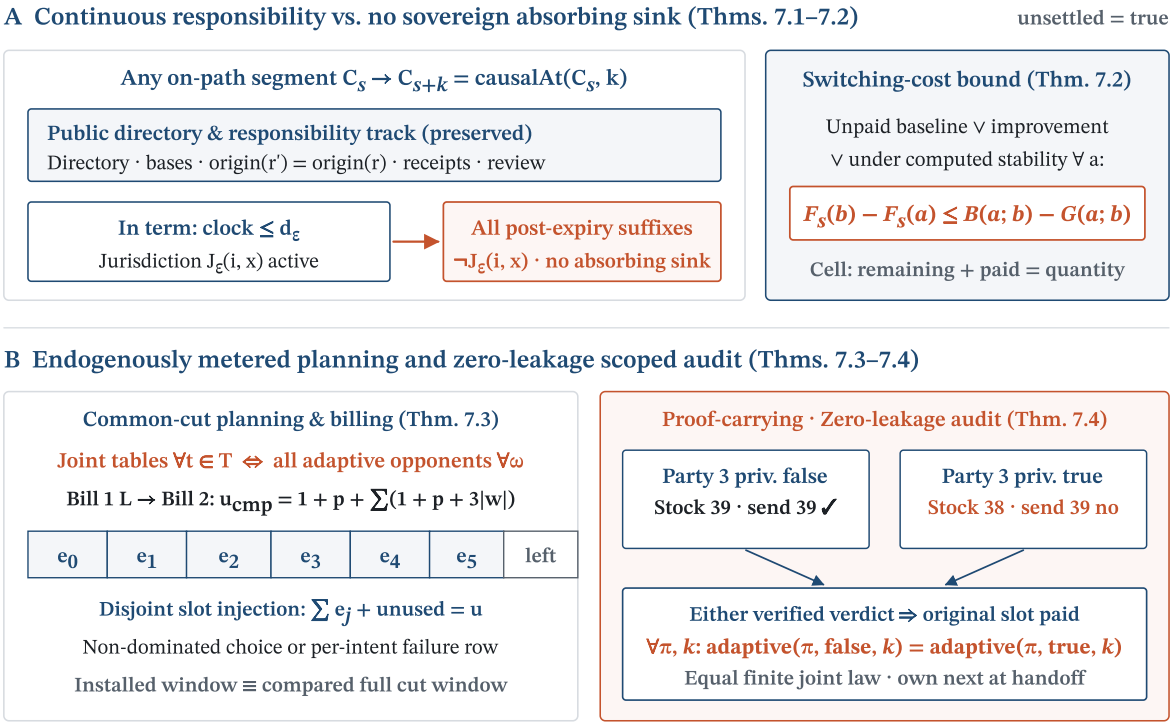


Figure 5 synthesizes Section 7: public offices carry expiry clocks, bounded switching costs, two-invoice planning bills, and dual-mask delivery proofs; public audit extracts zero bits about whether a citizen holds 38 or 39 units of private reserve, while each household locally reconstructs its exact root continuation.

8 External Boundaries, Intergenerational Upkeep, and Bookkeeping Discipline

An economic constitution that balances internal accounts while dumping ecological depletion, unpaid care, cross-jurisdictional harm, or decaying infrastructure onto outsiders and future generations is merely a charter for organized externalization.

> **Principle 7 (ECN7: Non-Externalization and Cross-Boundary Fiduciary Survival)**. Local financial closure, organizational dissolution, or zero internal net balance never extinguishes unsettled substantive obligations toward external counterparties, Earth-system boundaries, social reproduction, or future generations; pioneers' contributions to non-renewable commons or new frontiers settle up to Q_o and never convert into permanent exclusionary enclosure against latecomers.

8.1 Survival of Cross-Boundary External Liability After Local Closure

Shavell studied how limited attachable assets can weaken the deterrent effect of liability [25]. Rockström and colleagues proposed quantified safe and just Earth-system boundaries for climate, biosphere and water, among other domains [48]. In provenance-preserving double-entry accounting [37], local settlement and the discharge of external duties require their own evidence.

Theorem 8.1 (Survival of Cross-Boundary External Liability After Local Ledger Closure). There exists a countably infinite family of lawful institutional evolutions $\{e_{2,k}\}_{k\in\mathbb{N}}$ whose root, payment, and service occurrences are strictly injective in $k \in \mathbb{N}$, such that at each instance's post-settlement state:

$$\forall k \in \mathbb{N}, \qquad \mathrm{OpenLedger}_{\mathrm{loc}}(e_{2,k}) = \emptyset \ \ \wedge \ \ \Lambda_{\mathrm{ext}}(e_{2,k}) \neq \emptyset, \tag{23}$$

where $\mathrm{OpenLedger}_{\mathrm{loc}}(e_{2,k})$ is the local open obligation set and $\Lambda_{\mathrm{ext}}(e_{2,k})$ is the unsettled cross-boundary external liability set. The family disproves the implication from local closure to external discharge.

Proof. Fix a completed joint-settlement visit and its source index. Let $e_{2,k}$ be the kth actual current of its generated post-settlement continuation. Joint settlement produces an empty local ledger, and the exact root occurrence uses that same empty local support. The occurrence still generates external payment authority. Its certificate pairs the actual exit with the source-owned external liability read from that very root event, so $\Lambda_{\mathrm{ext}}(e_{2,k}) \neq \emptyset$. The current index is the initial continuation index plus k. The three recovery laws read it back from the root, payment, and service occurrences respectively, making each family injective. ■

Paired with Theorem 3.1(4), Theorem 8.1 seals both halves of corporate liability evasion: an extractive subsidiary or cross-border platform can neither point to balanced local books ($\mathrm{OpenLedger}_{\mathrm{loc}} = \emptyset$) to erase downstream watershed or labor liabilities ($\Lambda_{\mathrm{ext}} \neq \emptyset$) nor dissolve and reincorporate under empty immediate predecessors to claim debt-free birth ($\neg\,\mathrm{Birth} \wedge \mathrm{SameDebt}$).

8.2 Pioneers' Contributions Versus Latecomers' Equal Standing: From Terrestrial Commons to Outer Space

Cross-boundary accounting extends naturally across generations and newly opened domains, from deep seabeds and orbital spectrum to celestial bodies. Articles II, VI, and VIII of the 1967 Outer Space Treaty already proved in positive international law that facility jurisdiction, ongoing supervisory liability, and non-appropriation of the commons are cleanly separable: no state may appropriate celestial bodies by sovereignty, use, or occupation (Article II), while the state of registry retains jurisdiction and control over the registered object and its personnel (Article VIII) and the appropriate State Party authorizes and continually supervises non-governmental activities (Article VI) [17]. Combined with Rawls's and Buchanan's problem of latecomer consent [19; 21], Definition 1.1 and Theorems 4.2 and 8.1 establish a precise pioneer rule (Table 4): early builders may earn generous capped claims Q_o and exercise term-bound operational stewardship over facilities they maintain, **never hereditary enclosure over the frontier itself**.

Cross-Boundary Dimension	Standard Off-Ledger Externalization	Constitutional Hard Constraint (`ECN7`)	Formal Backing
Care & social reproduction	Treat caregiving and household upkeep as free background to subsidize trials	Unpaid current living/upkeep bundle immediately blocks next-round renewal	Thm. 4.1 (`paidNextTrial?`), Thm. 6.1
Earth-system boundaries	Offset irreversible carbon, pollution, or biosphere draws with cash profits	Separate physical unit accounts and ex-ante fiduciary quotas; no cross-unit netting	Thm. 5.1, Thm. 8.1
Cross-jurisdiction & shells	Net internal accounts or dissolve and re-register shells to shed external torts/debts	$\mathrm{OpenLedger}_{\mathrm{loc}} = \emptyset$ keeps $\Lambda_{\mathrm{ext}} \neq \emptyset$; reappearance is same-debt succession	Thm. 3.1(4), Thm. 8.1
Future generations & space	Turn early infrastructure outlay into permanent gatekeeping or deferred upkeep debt	Cap construction claims at Q_o; book upkeep net; preserve newcomers' equal standing	Def. 1.1, Thm. 4.2, Thm. 9.3

Table 4: Four constitutional boundaries preventing cost externalization onto care, ecosystems, external jurisdictions, and future generations.

9 Open Amendment, Anti-Capture, and World-Safe Regeneration

Every written rule eventually meets unanticipated technological shifts, resource shocks, or novel capture strategies. A constitution claiming eternal unamendability shatters into extra-legal sovereignty under stress [8; 16]; yet an amendment procedure that erases live debts or conjures phantom budgets out of renaming becomes an engine of default and usurpation.

> **Principle 8 (ECN8: Accounted Open Amendment, Fresh-Payment Standing, and World-Safe Regeneration)**. The constitution admits no unamendable terminal lock-in and preserves minimal coface amendment (`U8` / `self-U8`); yet stage transitions never refill exhausted obligation budgets, yesterday's payment receipts never substitute for today's basic security, and every periodic renewal must simultaneously preserve non-

contracting exit, capture escape, visible residual bearers, and cross-boundary source accountability.

9.1 Two-Layer Zero-Budget Squeeze and Amendment Without Phantom Refill

In classical AGM belief revision, inconsistent beliefs are contracted at zero physical cost [23]. Institutional ledgers carry live obligations and finite physical capacities: rule revision can reorganize procedures and register new residual rows, but cannot manufacture positive progress capacity out of relabeling.

Let row a carry natural-number progress budget $B(a) \geq 0$, inherited across stage entry $a \mapsto a'$ and initial writeback $a' \mapsto e$.

Theorem 9.1 (Two-Layer Zero-Budget Squeeze Closure and Unrefilled Amendment Boundary). At the first conservative stage successor and writeback of the same rooted exhausted ledger, progress budgets obey $0 \leq B(e) \leq B(a') \leq B(a)$; hence once $B(a) = 0$, every unreplenished stage transition and execution step satisfies

$$0 \leq B(e) \leq B(a') \leq B(a) = 0 \implies B(a') = B(e) = 0. \tag{24}$$

Simultaneously, an explicit three-row coface amendment witness (preserving inherited contribution, inherited obstruction, and newborn residual rows) produces a **valid institutional amendment answer** while its positive-budget subtype remains strictly empty, recording reorganization honestly without counterfeiting solvency.

Proof. Non-negativity gives $0 \leq B(e)$, execution monotonicity gives $B(e) \leq B(a')$, and stage-entry coverage gives $B(a') \leq B(a)$; substituting $B(a) = 0$ forces $B(a') = B(e) = 0$ in Equation (24). After exhausting contribution budget $1 \to 0$, minimal coface revision emits the three live rows with $B = 0$ alongside a valid residual response handler, inhabited without positive budget refill. ■

Thus when a pension reserve or municipal maintenance fund is exhausted ($B(a) = 0$), relabeling it into a new institutional stage cannot manufacture solvency ($B(a') = B(e) = 0$); lawful `U8` revision instead places the historical contribution, active obstruction, and newborn residual openly on the ledger awaiting genuine real-resource replenishment.

9.2 Strict Expansion of the Paid Authority Frontier and Six-Factor World-Safe Regeneration

How then does a living constitution renew itself across unbounded time without living off stale receipts or exporting hidden residuals? Proposition 9.2 and Theorem 9.3 close the dynamic architecture [1; 2; 3; 28; 25].

Let A_d denote the cumulative paid service-exit authority frontier at cycle d, K_d instantaneous outside-exit capacity, and $\prec_{\text{can}}$ the canonical constitutional strict order.

Proposition 9.2 (Strict Paid-Authority Frontier Expansion at Constant Instantaneous Capacity and Cofinal Self-Amendment). Along the registered renewal cycle $\{C_d\}_{d\in\mathbb{N}}$, instantaneous exit capacity stays $K_d = 1$ at every step while cumulative paid authorities expand by

strict inclusion, canonical constitutional order matches strict cycle order, and `self-U8` amendment remains cofinal:

$$A_0 \subsetneq A_1 \subsetneq A_2 \subsetneq \cdots, \qquad \forall d \in \mathbb{N},\ K_d = 1, \qquad \forall r, t \in \mathbb{N},\ (C_r \prec_{\text{can}} C_t \iff r < t). \tag{25}$$

Proof. Each step $d \to d+1$ executes a fresh payment occurrence whose visit index is injective, giving a newly born service-exit authority $\kappa(\gamma_d) \in A_{d+1} \setminus A_d$ and carrying the old frontier forward, hence $A_d \subsetneq A_{d+1}$, while maintaining one active uncaptured exit edge ($K_d = 1$) and yielding $C_r \prec_{\text{can}} C_t \iff r < t$ in Equation (25). For any horizon n, step $m = n + 1 > n$ realizes joint coface amendment, compatible future, and root-law continuation. ■

The six requirements for periodic regeneration come together in one question: **when society delivers its unconditional periodic basic payment p_d at depth d, can that single payment occurrence simultaneously secure internal freedom, capture repair, open amendment, and cross-boundary accountability without any pre-stored future script?**

Theorem 9.3 (Six-Factor World-Safe Regeneration Driven by One Same-Occurrence Periodic Payment). In the source-native canonical society process with zero pre-stored future tables, for **every** regeneration depth $d \in \mathbb{N}$ there exists a periodic basic payment occurrence p_d anchored to depth d's unique temporal root visit (with successive payments strictly distinct, $p_d \neq p_{d+1}$) simultaneously satisfying all six world-safe regeneration conditions $\mathcal{W}_d(p_d)$:

$$\forall d \in \mathbb{N}, \qquad \mathcal{W}_d(p_d) \equiv \text{Lin}_d(p_d) \wedge \text{Exit}_d(p_d) \wedge \text{Esc}_d(p_d) \wedge \text{Vis}_d(p_d) \wedge \text{Open}_d(p_d) \wedge \text{Ext}_d(p_d), \tag{26}$$

comprising:

1. **Same-lineage update and depth-zero amendment separation** (Lin_d): bounded repair retains the old law surface and takes the next temporal visit of the same root. At the separately witnessed policy break at depth $d = 0$, the ordinary and `self-U8` landings are inhabited and strictly distinct ($\ell_{\text{ord}} \neq \ell_{\text{rev}}$);
2. **Non-contracting ordinary exit capacity** (Exit_d): maintaining live outside-role exit at both source and successor ($1 \leq K_d \leq K_{d+1}$);
3. **Capture-kernel escape and target repair installation** (Esc_d): the captured configuration has empty feasible exit at p_d while installing the repaired target program at depth $d+1$;
4. **No invisible residual bearer** (Vis_d): the residual of this captured settlement retains its recorded bearer, with its activated consumer's next state identical to payment next state ($\text{next}(\text{consumer}) = \text{next}(p_d)$);
5. **Preserved `U8` amendment route without grounded terminal lock-in** (Open_d): the repaired successor carries a faithful coface map and an actual `U8` resumption. The designated tenure-repair image at this depth has an empty grounded-terminal type at landing `none`;
6. **Cross-boundary value and source-owned duty settlement** (Ext_d): separating external counterparty identity from the internal worker (counterparty $\neq$ worker) while preserving source-owned boundary obligations on p_d without off-ledger displacement.

Proof. Take the source-generated periodic step at depth d. Its two heterogeneous equalities identify the ordinary repair used by the captured settlement and by the repaired successor with the

repair on payment p_d. The original root compiles the bounded successor, retains its law surface, and moves to the root visit at depth $d + 1$. The original occurrence-depth injection separates the successive payments. Ordinary exit is inhabited at both visits with nondecreasing capacity. The captured exit type is empty, and the target service program equals the repair in the source disposition. The captured settlement retains its bearer and activated consumer, whose next state equals the payment's, and the target root supplies a responsibility consumer. The faithful coface map gives an actual `U8` resumption, and the designated tenure image has no grounded terminal at `none`. The external counterparty differs from the worker and receives the protected interest, with the exact settlement retaining source-owned responsibility. Separately, the depth-zero policy-break certificate gives both ordinary repair and `self-U8` at the same occurrence and separates their landings. ■

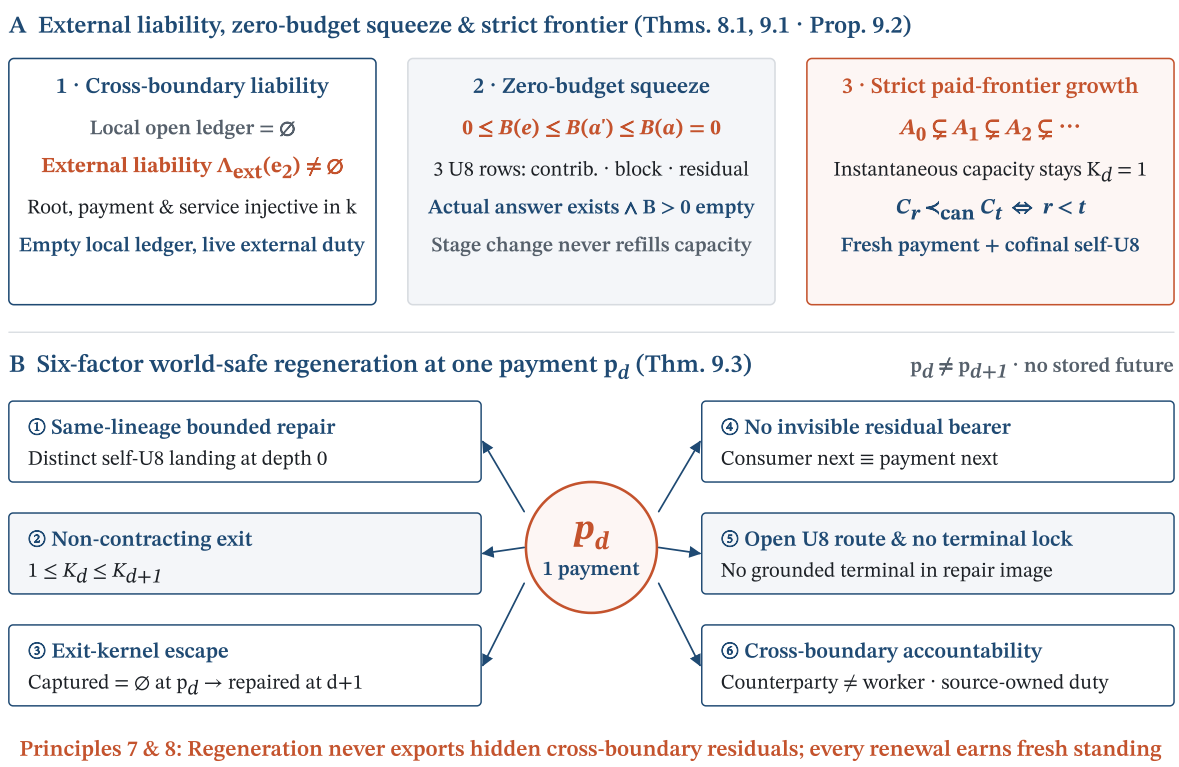


Figure 6 completes the constitutional arc: Panel A displays the three conservation guardrails (external duties and reappearing shells survive local closure, zero budgets never self-refill, and stale receipts never replace fresh payments), while Panel B shows how each honest periodic basic payment p_d simultaneously drives all six conditions of world-safe regeneration.

10 Five-Axis Synthesis, Institutional Boundaries, and Downstream Interfaces

10.1 Exact Theorem-Level Comparison Across Five Research Traditions

Table 5 synthesizes all fifty primary works into five interdisciplinary axes, contrasting established insights and open gaps against the exact formal theorems proved in this paper.

Research Axis & Primary Works	Core Mechanism Established in Prior Literature	Open Gap Left Unclosed	Formal Advance and Exact Boundary in This Paper
1. Constitutional political economy, public/private power & entitlement unbundling [6; 8; 9; 10; 16; 21; 29; 38; 47]	Separates claim-rights, privileges, and powers; exposes monopoly and corporate hierarchies as private government; interrogates latecomer consent and basic norms	Lacks a single-ledger formal criterion separating finite reward from perpetual rule and simultaneously barring sovereign exception sinks and private access domination	**Def. 1.1, Thm. 1.2, Thm. 2.1, Prop. 4.3, Thm. 7.1, Thm. 9.3**: Bars cross-row entitlement promotion; proves NoCapital $\Leftrightarrow$ NoAsymmetry; keeps unsettled duties live while expiring $\neg J_{\varepsilon}$ permanently; replaces extra-legal exception with coface `self-U8`
2. Capabilities, informational diagnosability & basic participation [5; 14; 18; 19; 22; 27; 35; 40; 41; 45; 50]	Shows goods/utility $\neq$ capabilities, links credible exit to voice and the power to say no, and debates cash UBI, basic services, and voluntary job guarantees	Stops at evaluative rankings or single-instrument advocacy without proving coarse repair obstruction, pre-capture demand balance, or anti-retaliation	**Thms. 2.1–2.2, Prop. 2.3, Thm. 3.1 (Table 2)**: Proves five-gate matching, Three Zeros, Two Ones, 2×2 repair obstruction, and $K + D^{\mathrm{orig}}_{\max} = \lvert C_{\mathrm{act}} \rvert$ invariant to re-quoting; excludes four retaliation retypings and debt-free reappearance
3. Uncertainty, firm contracts, capital coding & autonomous time [7; 11; 15; 20; 24; 26; 30; 39; 42; 44; 49]	Distinguishes uncertainty from risk; analyzes R&D appropriability, residual control, access power, four-module capital coding, unpaid care, and task displacement	Treats perpetual exclusionary control as the inevitable price of uncertainty-bearing or coordination; lacks a single-trace proof uniting method dividends and zero forced dispatch	**Thms. 4.1–4.2, Prop. 4.3, Thm. 6.1**: Adjudicates two-round out-of-menu discovery under current-living gate; caps all-slot receipts per origin at $R_o \leq Q_o$; proves whole-word $+1$ labor dividend, step-21 empty dispatch, and outside production
4. Commons peer production, dispersed knowledge, metered planning & clearing [12; 13; 31; 33; 34; 36; 37; 43]	Establishes modular commons production, polycentric provider replaceability, dispersed-knowledge limits, symmetric clearing, credit hierarchy, and double-entry algebra	Leaves physical upkeep unnetted in commons models, treats planning compute as costless in mechanism search, or masks unilateral non-delivery behind scalar net zero	**Thm. 5.1, Thms. 7.2–7.3 (Table 3)**: Proves $n + 1$ net fuel/upkeep dividend with per-good conservation; bounds switching net cost by $B - G$; equates finite joint tables with all adaptive opponents under two-invoice six-phase billing
5. Proof-carrying verification, contextual privacy, external liability & formal foundations [17; 23; 25; 28; 32; 46; 48; 1; 2; 3; 4]	Develops proof-carrying code, contextual privacy norms, judgment-proofness, outer-space non-appropriation, Earth-system boundaries, and belief revision	Anti-fraud enforcement drifts toward panopticism; belief contraction ignores live debt budgets; local closure or shell re-registration is conflated with immunity	**Thms. 3.1(4), 7.4, 8.1, 9.1, Prop. 9.2, Thm. 9.3**: Proves dual-mask soundness with adaptive/timed zero leakage and `rebuild` own-frame root next; preserves external and reappearing-shell debts; proves zero-budget squeeze and six-factor regeneration

Table 5: Exact theorem-level advances and scope boundaries across five primary research traditions.

10.2 Conclusion and Downstream Treatise Interfaces

An economic constitution suited to rapidly expanding possibilities requires neither selfless saints nor an omniscient sovereign planner or panoptic surveillance apparatus. It rests on four verifiable constitutional disciplines: first, **decoupling survival standing from prior output proof**, measuring substantive freedom by five-gate outside exit, original demand conservation, connected service portfolios, and structural anti-retaliation (Sections 2–3); second, **pricing open-ended exploration, risk, waiting, and upkeep transparently under a current-living renewal gate while capping cumulative receipts per issuance origin at Q_o** so finite contributions never harden into permanent rule (Section 4); third, **releasing settled non-rival methods into zero-fee public reuse and returning net labor savings as autonomous time while clearing rival materials and credit legs item by item** (Sections 5–6); and fourth, **keeping public-function duties continuous while expiring office jurisdiction on schedule, billing planning across two invoices, requiring proof-carrying zero-leakage public audit that preserves each household's exact private `rebuild` continuation, and driving six-factor world-safe regeneration from each fresh periodic basic payment** (Sections 7–9).

The formal results give concrete ways to maintain the four entitlements in their respective domains. The supplement and companion papers carry the downstream implementations. (i) multi-channel monetary issuance, concurrent double-spend prevention, and challenge protocols are specified as an engineering contract in Supplement Appendix D.1 rather than claimed here as a deployed currency trial; (ii) finer-grained scoped projection over local clock-150 recursive production menus and cryptographic side-channel hardening are specified in Supplement Appendix D.2; and (iii) universal minimal-permission abolishability and zero-prior-probability risk pricing are proved in the companion treatise on capitalism [4], while multi-provider public-function concurrency, full-scale public production networks, accounted institutional transitions, and enforceable inter-polity peace are developed in their dedicated treatises. Delivering razor-sharp machine-verified watersheds while marking downstream engineering interfaces openly is how this constitution practices its own accounting ethic.

Sources and Proofs

All eighteen named formal results in this paper (Theorems 1.2, 2.1, 2.2, Proposition 2.3, Theorems 3.1, 4.1, 4.2, Proposition 4.3, Theorems 5.1, 6.1, 7.1, 7.2, 7.3, 7.4, 8.1, 9.1, Proposition 9.2, and Theorem 9.3) have corresponding constructions, counterexamples and consumers verified in the standard kernel of Lean 4 [46] (relying solely on `propext`, `Quot.sound`, and `Classical.choice`, with `unsafe = 0`, `partial = 0`, and `sorry = 0`). The third-edition source map gives each claim's original epoch, public owner, consumers, and complete type. It links the original economic and social proof snapshots and the finite-product proof. The reproduction guide gives the checking commands.

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